Show that f(x) = 4x + 1 grows by the same amount over every interval of length 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q1/5

801

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

f(x + 2) − f(x) = 4(x + 2) + 1 − (4x + 1) = 8, for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q1/5

Show that g(x) = 3^x grows by the same factor over every interval of length 1.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q2/5

802

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

g(x + 1) ÷ g(x)
= 3^(x + 1) ÷ 3^x
= 3, for every x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q2/5

Which function grows by equal factors over equal intervals?

A. $y = 2x^2$

B. $y = x + 5$

C. $y = 5x + 2$

D. $y = 5(2)^x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q3/5

803

QUESTION ID

S26-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y = 5(2)^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q3/5

True or false?

The linear function $f(x) = mx + b$ changes by $m(x_2 - x_1)$ from x_1 to x_2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q4/5

804

QUESTION ID

S26-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q4/5

True or false?

$y = x^2$ grows by equal differences over equal intervals.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q5/5

805

QUESTION ID

S26-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.1.a • Set 26 • FORWARD 1 • Q5/5

SET 26 • 8.F.A.3

FORWARD 2

Use factoring to find the zeros and maximum of a quadratic function • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a

ANSWER KEY Reduced copy of the question page, with its answer

A ball's height is $h(t) = -16t^2 + 32t = -16t(t - 2)$ feet after t seconds.

Use the factored form to find when the ball is at height 0.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q1/5

807

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

t = 0 and t = 2 seconds

GRADING NOTE Both times are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q1/5

A ball's height is $h(t) = -16t(t - 2)$ feet after t seconds. Its greatest height is halfway between the zeros.

When does it reach its greatest height, and what is that height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q2/5

808

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

At t = 1 second; 16 feet

GRADING NOTE Both the time and the height are required.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q2/5

$f(x) = x^2 - 6x + 8 = (x - 2)(x - 4)$

What are the zeros of f?

A. 2 and 4

B. −2 and −4

C. 8 only

D. 6 and 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q3/5

809

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q3/5

True or false?

The graph of a quadratic function is symmetric about a vertical line through its vertex.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q4/5

QUESTION ID

S26-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q4/5

True or false?

The zeros of f(x) = (x − 3)(x + 5) are 3 and 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q5/5

811

QUESTION ID

S26-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The zeros are 3 and −5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S26-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.8.a • Set 26 • FORWARD 2 • Q5/5

SET 27 • 8.F.B.4 • FUNCTIONS

Find the rate of change and initial value from two points, a table, or a graph

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Find a constant rate from a table	**7.RP.A.2.b**
BACKWARD 2	Subtract integers	**7.NS.A.1.c**
FORWARD 1	Construct linear and exponential functions from two points	**HSF-LE.A.2**
FORWARD 2	Write arithmetic and geometric sequences recursively and explicit..	**HSF-BF.A.2**

SET 27 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

A linear function passes through (2, 7) and (5, 16).

What are its rate of change and its initial value?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q1/5

814

QUESTION ID

S27-M-Q1

TYPE

Short answer

CORRECT ANSWER

Rate of change 3; initial value 1

GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q1/5

The table shows a linear function.

What are its rate of change and its initial value?

x	1	3	5
y	10	16	22

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q2/5

815

QUESTION ID

S27-M-Q2

TYPE

Short answer

CORRECT ANSWER

Rate of change 3; initial value 7

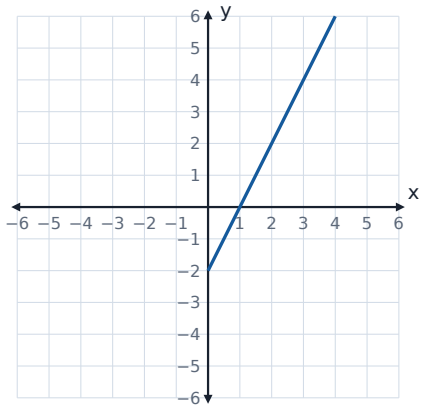
GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q2/5

The graph shows a linear function.

What are its rate of change and its initial value?



A. Rate 4, initial value −2

B. Rate −2, initial value 2

C. Rate 2, initial value −2

D. Rate 2, initial value 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q3/5

816

QUESTION ID

S27-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Rate 2, initial value −2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q3/5

A line passes through (−1, 8) and (3, 0).

Write its equation in the form y = mx + b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q4/5

817

QUESTION ID

S27-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = −2x + 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q4/5

True or false?

A linear function with f(0) = 5 and f(2) = 1 has a rate of change of −2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q5/5

818

QUESTION ID

S27-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 27 • MAIN • Q5/5

SET 27 • 8.F.B.4

BACKWARD 1

Find a constant rate from a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant rate (y per x)?

x	3	6	9
y	12	24	36

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q1/5

820

QUESTION ID

S27-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q1/5

The table shows a proportional relationship.

What is the constant rate (y per x)?

Hours	2	5	8
Miles	7	17.5	28

Rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q2/5

821

QUESTION ID

S27-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3.5 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q2/5

A proportional table contains the pair (4, 10).

What is the constant rate (y per x)?

A. 0.4

B. 40

C. 2.5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q3/5

822

QUESTION ID

S27-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q3/5

True or false?

A proportional table with the pairs (2, 1) and (6, 3) has a constant rate of 0.5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q4/5

823

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q4/5

True or false?

A proportional table with the pairs (5, 15) and (8, 24) has a constant rate of 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q5/5

824

QUESTION ID

S27-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 8.F.B.4

BACKWARD 2

Subtract integers • Grade 7 • The Number System • 7.NS.A.1.c

Subtract.

$6 - (-2)$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q1/5

826

QUESTION ID

S27-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q1/5

Subtract.

$-3 - 5$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q2/5

827

QUESTION ID

S27-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q2/5

Subtract.

$-4 - (-9)$

A. 5

B. 13

C. -13

D. -5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q3/5

828

QUESTION ID

S27-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q3/5

True or false?

$2 - 7 = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q4/5

829

QUESTION ID

S27-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q4/5

True or false?

$-6 - (-6) = -12$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q5/5

QUESTION ID

S27-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 27 • BACKWARD 2 • Q5/5

SET 27 • 8.F.B.4

FORWARD 1

Construct linear and exponential functions from two points • High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2

Write the linear function whose graph passes through (1, 4) and (3, 10).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q1/5

832

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3x + 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q1

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q1/5

Write the exponential function f(x) = a · b^x whose graph passes through (0, 5) and (1, 15).

f(x) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q2

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q2/5

833

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5 · 3^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S27-F1-Q2**

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q2/5

Which exponential function passes through (0, 2) and (2, 18)?

A. f(x) = 8x + 2

B. f(x) = 2 · 3^x

C. f(x) = 18 · 2^x

D. f(x) = 2 · 9^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q3/5

834

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. f(x) = 2 · 3^x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q3

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q3/5

True or false?

The linear function through (0, −1) and (4, 7) is f(x) = 2x − 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q4/5

835

QUESTION ID

S27-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q4

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q4/5

True or false?

The exponential function through (0, 4) and (1, 2) is f(x) = 4 · 2^x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q5/5

836

QUESTION ID

S27-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The function is f(x) = 4 · (0.5)^x.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F1-Q5

High School (Grade 9 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.A.2 • Set 27 • FORWARD 1 • Q5/5

SET 27 • 8.F.B.4

FORWARD 2

Write arithmetic and geometric sequences recursively and explicitly • High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2

Write an explicit formula for the arithmetic sequence 7, 11, 15, ..., starting with n = 1.

a(n) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q1/5

838

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4n + 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q1

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q1/5

Write a recursive formula for the sequence 7, 11, 15, ...

Formula: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q2/5

839

QUESTION ID

S27-F2-Q2

TYPE

Short answer

CORRECT ANSWER

a(1) = 7;

a(n) = a(n − 1) + 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q2/5

Which explicit formula gives 3, 6, 12, 24, ..., starting with n = 1?

A. a(n) = 2 · 3^(n − 1)

B. a(n) = 3 · 2^(n − 1)

C. a(n) = 3 + 2n

D. a(n) = 3n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q3/5

840

QUESTION ID

S27-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. a(n) = 3 · 2^(n − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q3

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q3/5

True or false?

An arithmetic sequence is a linear function whose domain is the counting numbers.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q4/5

841

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q4

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q4/5

True or false?

The sequence 2, 5, 8, 11, ... has a common ratio of 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q5/5

842

QUESTION ID
S27-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It has a common difference of 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S27-F2-Q5

High School (Grade 10 in this bank) • Functions: Building Functions • HSF-BF.A.2 • Set 27 • FORWARD 2 • Q5/5

SET 28 • 8.F.B.4 • FUNCTIONS

Construct a linear function to model a situation

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Write an equation for a situation	**6.EE.C.9**
BACKWARD 2	Solve a two-step equation from a model	**7.EE.B.4.a**
FORWARD 1	Write a function for a situation and use it to answer a question	**HSF-BF.A.1.a**
FORWARD 2	Rearrange a formula to solve for a chosen quantity	**HSA-CED.A.4**

SET 28 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

A plumber charges $65 for a visit plus $40 per hour.

Write a function for the cost C after h hours.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q1/5

845

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

40h + 65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q1/5

A candle is 30 cm tall and burns down 2.5 cm each hour.

Write a function for its height h after t hours.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q2/5

846

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

30 − 2.5t

GRADING NOTE −2.5t + 30 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q2/5

A phone plan costs $20 per month after a $50 activation fee.

Which function gives the total cost C after m months?

A. C = 20m − 50

B. C = 20m + 50

C. C = 50m + 20

D. C = 70m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q3/5

847

QUESTION ID

S28-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. C = 20m + 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q3/5

The cost of a taxi ride is a linear function of the miles m. A 3-mile ride costs $11 and a 7-mile ride costs $21.

Write the function.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q4/5

848

QUESTION ID

S28-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.5m + 3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q4/5

True or false?

A tank holds 500 liters and drains 20 liters each minute. The function V = 500 − 20t gives the liters V after t minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q5/5

849

QUESTION ID

S28-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 28 • MAIN • Q5/5

SET 28 • 8.F.B.4

BACKWARD 1

Write an equation for a situation • Grade 6 • Expressions & Equations • 6.EE.C.9

Tickets cost $9 each.

Write an equation for the cost c of t tickets.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

851

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

c = 9t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q1/5

Mia has $15 and saves $5 each week.

Write an equation for her savings s after w weeks.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

852

QUESTION ID

S28-B1-Q2

TYPE

Short answer

CORRECT ANSWER

s = 5w + 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q2/5

A pool holds 100 gallons and fills at 10 gallons per minute.

Which equation gives the gallons g after m minutes?

A. g = 110m

B. g = 100m + 10

C. g = 10m + 100

D. g = 10m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

853

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. g = 10m + 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q3/5

True or false?

Apples cost $3 per pound. The equation c = 3p gives the cost c of p pounds.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

854

QUESTION ID

S28-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q4/5

True or false?

A game costs a $10 fee plus $2 per game. The equation c = 10g + 2 gives the cost c of g games.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

855

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The equation is c = 2g + 10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 8.F.B.4

BACKWARD 2

Solve a two-step equation from a model • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for h.

40h + 65 = 185

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q1/5

857

QUESTION ID

S28-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q1/5

Solve for t.

$30 - 2.5t = 10$

t = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q2/5

858

QUESTION ID

S28-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q2/5

Solve for m.

20m + 50 = 170

A. m = 3

B. m = 11

C. m = 8.5

D. m = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q3/5

859

QUESTION ID

S28-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. m = 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q3/5

True or false?

t = 4 is the solution of 500 − 20t = 420.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q4/5

860

QUESTION ID

S28-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q4/5

True or false?

h = 5 is the solution of 12h + 8 = 50.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q5/5

861

QUESTION ID

S28-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is h = 3.5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 28 • BACKWARD 2 • Q5/5

SET 28 • 8.F.B.4

FORWARD 1

Write a function for a situation and use it to answer a question • High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a

A gym charges $25 per month, but the first month is free.

Write a function for the total cost C after m months, where m ≥ 1.

C(m) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q1/5

863

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

25(m − 1)

GRADING NOTE 25m − 25 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q1

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q1/5

A savings account starts with $300, and $40 is added each week.

Write a function S(w), then find when the balance reaches $700.

Function and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q2

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q2/5

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

S(w) = 40w + 300; 10 weeks

GRADING NOTE Grade both: the function and 10 weeks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S28-F1-Q2**

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q2/5

A pool is 4 feet deep and drains 0.25 foot each hour.

Which function gives the depth after t hours, and when is the pool empty?

A. d(t) = 4 − 0.25t; after 16 hours

B. d(t) = 0.25t + 4; after 16 hours

C. d(t) = 4t − 0.25; after 1 hour

D. d(t) = 4 − 0.25t; after 4 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q3/5

865

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. d(t) = 4 − 0.25t; after 16 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q3

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q3/5

True or false?

A rental costs $30 plus $0.20 per mile. The function C(m) = 0.2m + 30 gives the cost for m miles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q4/5

866

QUESTION ID

S28-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q4

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q4/5

True or false?

A balance of $200 decreases by $15 each month. The function B(m) = 15m + 200 gives the balance after m months.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q5/5

867

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The function is B(m) = 200 − 15m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F1-Q5

High School (Grade 9 in this bank) • Functions: Building Functions • HSF-BF.A.1.a • Set 28 • FORWARD 1 • Q5/5

SET 28 • 8.F.B.4

FORWARD 2

Rearrange a formula to solve for a chosen quantity • High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4

Solve C = 40h + 65 for h.

h = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q1/5

869

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(C − 65)/40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q1

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q1/5

Solve $F = \frac{9}{5}C + 32$ for C.

C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q2/5

870

QUESTION ID

S28-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(5/9)(F − 32)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q2

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q2/5

Solve $y = mx + b$ for b.

A. $b = \frac{y}{mx}$

B. $b = mx - y$

C. $b = y + mx$

D. $b = y - mx$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q3/5

871

QUESTION ID

S28-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. b = y − mx

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q3

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q3/5

True or false?

For $t \neq 0$, solving $d = rt$ for r gives $r = \frac{d}{t}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q4/5

872

QUESTION ID

S28-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q4

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q4/5

True or false?

For $l \neq 0$, solving $A = lw$ for w gives $w = A - l$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q5/5

873

QUESTION ID
S28-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE $w = A/l$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S28-F2-Q5

High School (Grade 10 in this bank) • Algebra: Creating Equations • HSA-CED.A.4 • Set 28 • FORWARD 2 • Q5/5

SET 29 • 8.F.B.4 • FUNCTIONS

Interpret the rate of change and initial value of a linear function in context

MAIN	Grade 8 main question	**8.F.B.4**
BACKWARD 1	Explain points on the graph of a proportional relationship	**7.RP.A.2.d**
BACKWARD 2	Identify the independent and dependent variables	**6.EE.C.9**
FORWARD 1	Interpret the slope and intercept of a linear model of data	**HSS-ID.C.7**
FORWARD 2	Interpret the parameters of linear and exponential functions	**HSF-LE.B.5**

SET 29 • 8.F.B.4

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.4

Renting a bike costs C = 4h + 10 dollars for h hours.

What do the 4 and the 10 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q1/5

876

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

4: the cost goes up $4 for each hour. 10: a $10 fee (the cost at 0 hours).

GRADING NOTE Grade both numbers.

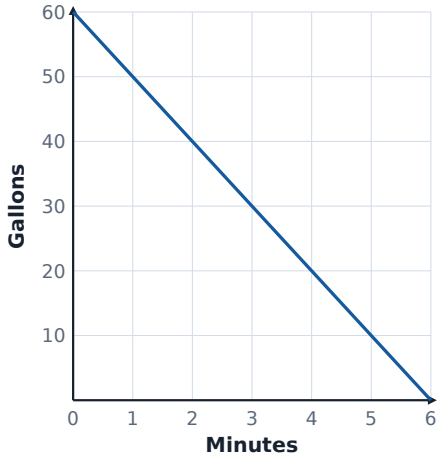
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q1

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q1/5

The graph shows the water in a tank.

What do the rate of change and the initial value mean?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q2/5

877

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

The tank starts with 60 gallons (initial value) and drains 10 gallons per minute (rate of change −10).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q2

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q2/5

A plant's height is h = 0.5w + 12 centimeters after w weeks.

What does 0.5 mean?

A. The plant grows for 0.5 week.

B. The plant grows 0.5 cm each week.

C. The plant was 0.5 cm tall at the start.

D. The plant grows 12 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q3/5

878

QUESTION ID

S29-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The plant grows 0.5 cm each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q3

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q3/5

The table shows the money in a savings account.

What does the rate of change mean?

Weeks	0	4	8
Dollars	150	210	270

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q4/5

879

QUESTION ID

S29-M-Q4

TYPE

Short answer

CORRECT ANSWER

$15 is added each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q4/5

True or false?

For C = 4h + 10, the initial value means the bike costs $4 before any hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q5/5

880

QUESTION ID

S29-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The initial value is $10.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-M-Q5

Grade 8 • Functions • 8.F.B.4 • Set 29 • MAIN • Q5/5

SET 29 • 8.F.B.4

BACKWARD 1

Explain points on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

A proportional graph of gallons (x) and miles (y) passes through (1, 28).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q1/5

882

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

The car travels 28 miles on 1 gallon.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q1/5

A proportional graph shows hours worked (x) and pay (y).

What does the point (0, 0) mean?

A. 0 hours of work earn $0.

B. The worker starts with $0 per hour.

C. The pay is $0 per hour.

D. There is no graph.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q2/5

883

QUESTION ID

S29-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 0 hours of work earn $0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q2/5

A proportional graph of minutes (x) and laps (y) passes through (6, 3).

What does this point mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q3/5

884

QUESTION ID

S29-B1-Q3

TYPE

Short answer

CORRECT ANSWER

3 laps are swum in 6 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q3/5

True or false?

On the graph of a proportional relationship, the point (1, r) shows the unit rate r.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q4/5

885

QUESTION ID

S29-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q4/5

True or false?

On a graph of tickets (x) and cost (y), the point (2, 18) means each ticket costs $18.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q5/5

886

QUESTION ID
S29-B1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It means 2 tickets cost $18 ($9 each).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 8.F.B.4

BACKWARD 2

Identify the independent and dependent variables • Grade 6 • Expressions & Equations • 6.EE.C.9

In C = 4h + 10, which variable is the independent variable?

Variable: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q1/5

888

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

h

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q1/5

"Pay depends on the hours worked." Which quantity is the dependent variable?

Quantity: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q2/5

889

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Pay

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q2/5

In d = 60t, what is d?

A. The rate

B. The independent variable

C. A constant

D. The dependent variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q3/5

890

QUESTION ID

S29-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The dependent variable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q3/5

True or false?

In a table of hours worked and pay, the hours are usually the independent variable.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q4/5

891

QUESTION ID

S29-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q4/5

True or false?

In y = 3x + 2, x depends on y.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q5/5

892

QUESTION ID

S29-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 29 • BACKWARD 2 • Q5/5

SET 29 • 8.F.B.4

FORWARD 1

Interpret the slope and intercept of a linear model of data • High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7

A line of fit predicts a test score y = 4.5x + 60 for x hours of study.

What does 4.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q1/5

894

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Each extra hour of study goes with about 4.5 more points.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q1

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q1/5

A line of fit predicts a test score y = 4.5x + 60 for x hours of study.

What does 60 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q2/5

895

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The predicted score for 0 hours of study.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q2

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q2/5

A model gives a used car's value as $y = 18{,}000 - 1{,}500x$ dollars after x years.

What does $-1{,}500$ mean?

A. The car is 1,500 years old.

B. The value drops about $1,500 each year.

C. The car costs $1,500.

D. The value rises $1,500 each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q3/5

896

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The value drops about $1,500 each year.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q3

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q3/5

True or false?

A model gives a child's height as y = 2.1x + 30 cm at age x months. The intercept 30 is the predicted height at 0 months.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q4/5

QUESTION ID

S29-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q4

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q4/5

True or false?

A slope of −3 in a linear model means y increases by 3 for each unit of x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q5/5

898

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F1-Q5

High School (Grade 9 in this bank) • Statistics & Probability: Interpreting Categorical & Quantitative Data • HSS-ID.C.7 • Set 29 • FORWARD 1 • Q5/5

SET 29 • 8.F.B.4

FORWARD 2

Interpret the parameters of linear and exponential functions • High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5

ANSWER KEY Reduced copy of the question page, with its answer

P(t) = 2,000(1.05)^t gives an account balance after t years.

What do 2,000 and 1.05 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q1/5

900

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2,000 is the starting amount; 1.05 means the balance grows 5% each year.

GRADING NOTE Grade both.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q1

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q1/5

V(t) = 25,000(0.85)^t gives a car's value after t years.

What is the yearly percent change?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q2/5

901

QUESTION ID

S29-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A 15% decrease each year

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q2

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q2/5

A phone plan costs C(m) = 0.12m + 9.99 dollars for m minutes.

What does 9.99 represent?

A. A fixed monthly charge

B. The number of minutes

C. The total cost

D. The cost per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q3/5

902

QUESTION ID

S29-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A fixed monthly charge

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q3

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q3/5

True or false?

In f(t) = 100(2)^t, the quantity doubles each time t increases by 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q4/5

903

QUESTION ID

S29-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q4

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q4/5

True or false?

In f(t) = 50(0.9)^t, the quantity grows by 90% each period.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q5/5

904

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It decreases by 10% each period.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S29-F2-Q5

High School (Grade 10 in this bank) • Functions: Linear, Quadratic & Exponential Models • HSF-LE.B.5 • Set 29 • FORWARD 2 • Q5/5

SET 30 • 8.F.B.5 • FUNCTIONS

Describe the relationship shown by a graph

MAIN	Grade 8 main question	**8.F.B.5**
BACKWARD 1	Interpret points on a graph of a situation	**5.G.A.2**
BACKWARD 2	Compare speeds as unit rates	**6.RP.A.3.b**
FORWARD 1	Interpret the key features of a graph in context	**HSF-IF.B.4**
FORWARD 2	Graph and evaluate piecewise-defined functions	**HSF-IF.C.7.b**

SET 30 • 8.F.B.5

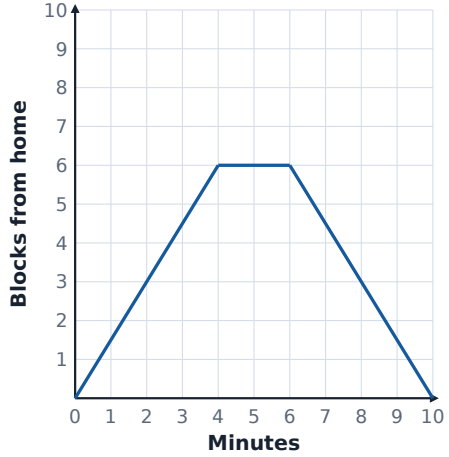
MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.5

The graph shows Ana's walk.

Describe what happens from 4 to 6 minutes.

Description: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q1/5

907

QUESTION ID

S30-M-Q1

TYPE

Short answer

CORRECT ANSWER

She stays 6 blocks from home; she stops.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q1/5

The graph shows Ana's walk.

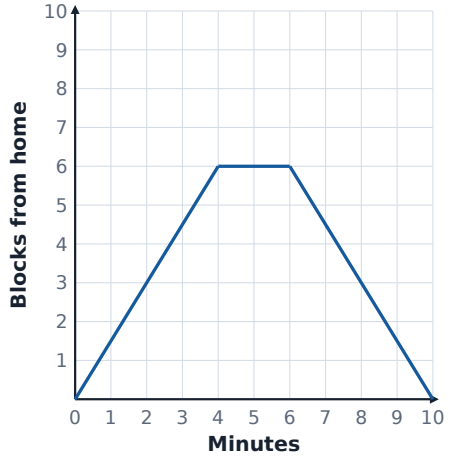
During which interval is Ana moving toward home?

A. 0 to 4 minutes

B. 6 to 10 minutes

C. 4 to 6 minutes

D. 0 to 6 minutes



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q2/5

908

QUESTION ID

S30-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 6 to 10 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

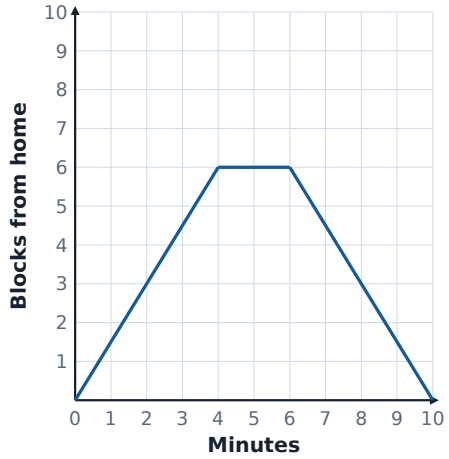
ID S30-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q2/5

The graph shows Ana's walk.

Is the relationship linear or nonlinear over the whole walk? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q3/5

909

QUESTION ID

S30-M-Q3

TYPE

Short answer

CORRECT ANSWER

Nonlinear. The graph is not one straight line; the rate of change is different on different parts.

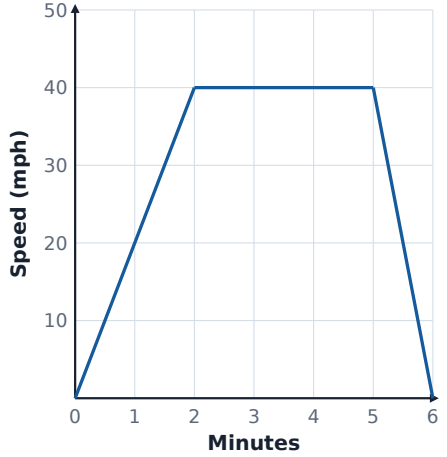
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q3/5

The graph shows a car's speed.

Describe how the speed changes over the 6 minutes.



Description: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q4/5

910

QUESTION ID

S30-M-Q4

TYPE

Short answer

CORRECT ANSWER

The car speeds up for 2 minutes, travels at a steady 40 mph until minute 5, then slows to a stop.

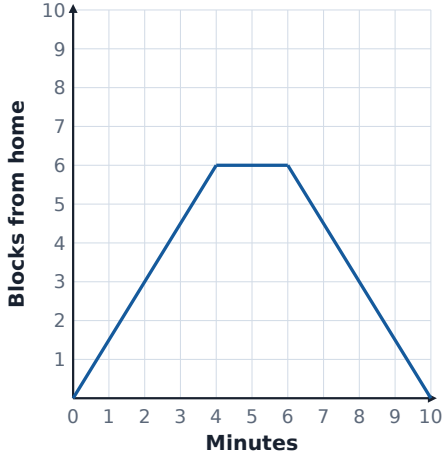
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q4/5

True or false?

In the graph of Ana's walk, she walks faster going home than going away from home.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q5/5

911

QUESTION ID

S30-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Both trips cover 6 blocks in 4 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 30 • MAIN • Q5/5

SET 30 • 8.F.B.5

BACKWARD 1

Interpret points on a graph of a situation • Grade 5 • Geometry • 5.G.A.2

A graph shows minutes (x) and blocks from home (y). What does the point (3, 5) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q1/5

913

QUESTION ID

S30-B1-Q1

TYPE

Short answer

CORRECT ANSWER

After 3 minutes, the walker is 5 blocks from home.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B1-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q1/5

A graph shows days (x) and plant height in centimeters (y). What does the point (10, 4) mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q2/5

914

QUESTION ID

S30-B1-Q2

TYPE

Short answer

CORRECT ANSWER

After 10 days, the plant is 4 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q2/5

A graph shows hours (x) and miles driven (y).

Which point shows 120 miles after 2 hours?

A. (2, 120)

B. (60, 2)

C. (2, 60)

D. (120, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q3/5

915

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (2, 120)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q3/5

True or false?

On a graph of time (x) and distance (y), the point (0, 0) can mean the start of a trip.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q4/5

916

QUESTION ID

S30-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q4/5

True or false?

The point (4, 1) on a graph of hours (x) and miles (y) means 4 miles in 1 hour.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q5/5

917

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 8.F.B.5

BACKWARD 2

Compare speeds as unit rates • Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b

Ana walks 6 blocks in 4 minutes at a steady pace.

What is her speed in blocks per minute?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q1/5

919

QUESTION ID

S30-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1.5 blocks per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q1/5

Which is faster: 12 miles in 3 hours, or 10 miles in 2 hours?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q2/5

920

QUESTION ID

S30-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10 miles in 2 hours (5 mph, compared with 4 mph)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-B2-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q2/5

Which rate is the greatest?

A. 60 miles in 1 hour

B. 100 miles in 2 hours

C. 20 miles in 0.5 hour

D. 150 miles in 3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q3/5

921

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 miles in 1 hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q3/5

True or false?

Walking 3 miles in 1 hour is faster than walking 5 miles in 2 hours.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q4/5

922

QUESTION ID

S30-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q4/5

True or false?

Running 8 km in 40 minutes is the same rate as running 2 km in 20 minutes.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q5/5

923

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 8.F.B.5

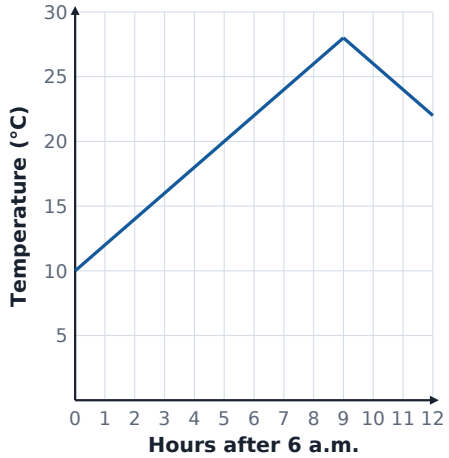
FORWARD 1

Interpret the key features of a graph in context • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

The graph shows the temperature during a day.

When is the temperature highest, and what is it?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q1/5

925

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9 hours after 6 a.m. (3 p.m.); 28°C

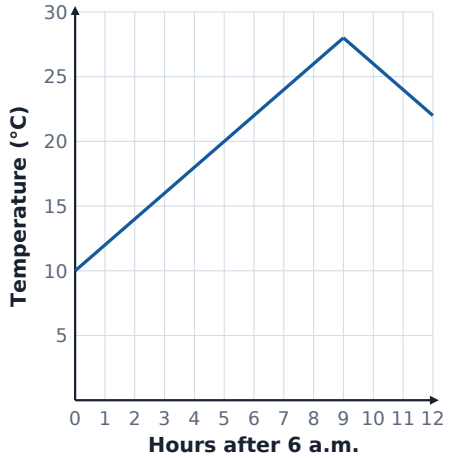
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q1/5

The graph shows the temperature during a day.

On what interval is the temperature increasing?

Interval: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q2/5

926

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

From 0 to 9 hours after 6 a.m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q2/5

The graph shows the temperature during a day.

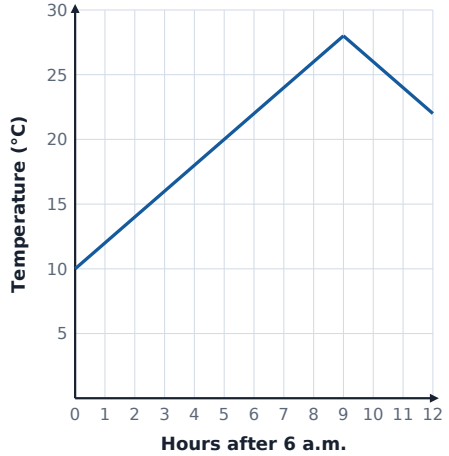
What does the y-intercept mean?

A. It was 28°C at 3 p.m.

B. It was 10°C at 6 a.m.

C. It was 22°C at 6 p.m.

D. The temperature rose for 9 hours.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q3/5

927

QUESTION ID

S30-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. It was 10°C at 6 a.m.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q3

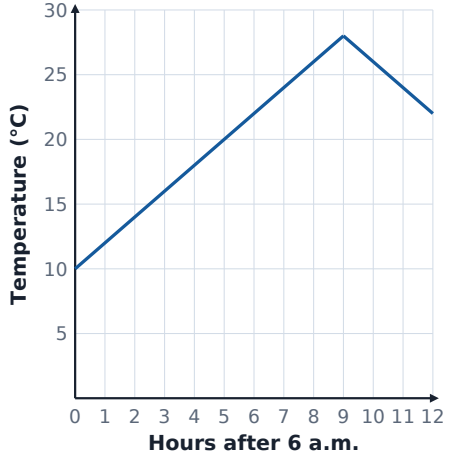
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q3/5

True or false?

In the graph, the temperature is decreasing from 9 to 12 hours after 6 a.m.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q4/5

928

QUESTION ID

S30-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q4/5

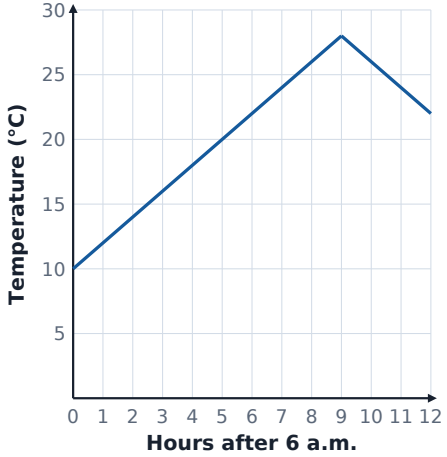
ANSWER KEY Reduced copy of the question page, with its answer

True or false?

The graph shows that the temperature never dropped below 22°C.

A. True

B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q5/5

929

QUESTION ID

S30-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It started at 10°C.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 30 • FORWARD 1 • Q5/5

SET 30 • 8.F.B.5

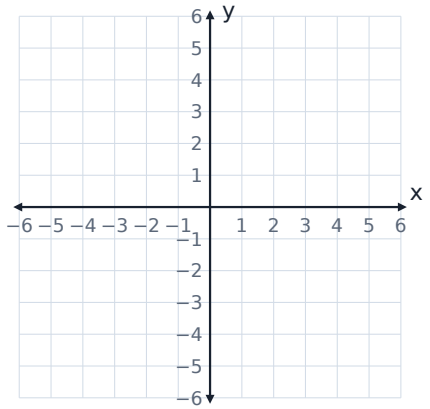
FORWARD 2

Graph and evaluate piecewise-defined functions • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b

Graph the function on the coordinate plane.

$f(x) = x + 2$ for $x < 0$

$f(x) = 2$ for $x \geq 0$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q1/5

931

QUESTION ID

S30-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The line $y = x + 2$ for $x < 0$ (an open end at (0, 2) is acceptable) and the horizontal line $y = 2$ for $x \geq 0$, with a closed point at (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q1/5

$f(x) = 3$ for $x \leq 2$, and $f(x) = x + 1$ for $x > 2$.

Find $f(5)$.

$f(5) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q2/5

932

QUESTION ID

S30-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q2/5

A taxi charges $3 for the first mile and $2 for each mile after that.

Which function gives the cost for m miles?

A. C(m) = 2m + 3 for all m

B. C(m) = 3m for all m

C. C(m) = 3 for 0 < m ≤ 1; C(m) = 3 + 2(m − 1) for m > 1

D. C(m) = 3 + 2m for m ≤ 1 only

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q3/5

933

QUESTION ID

S30-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. C(m) = 3 for 0 < m ≤ 1; C(m) = 3 + 2(m − 1) for m > 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q3/5

True or false?

For $f(x) = |x|$, $f(-4) = 4$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q4/5

934

QUESTION ID

S30-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q4/5

True or false?

A piecewise-defined function can never have an unbroken graph.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q5/5

935

QUESTION ID

S30-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE For example, f(x) = |x| is piecewise-defined and unbroken.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S30-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.b • Set 30 • FORWARD 2 • Q5/5

SET 31 • 8.F.B.5 • FUNCTIONS

Sketch a graph from a description of a relationship

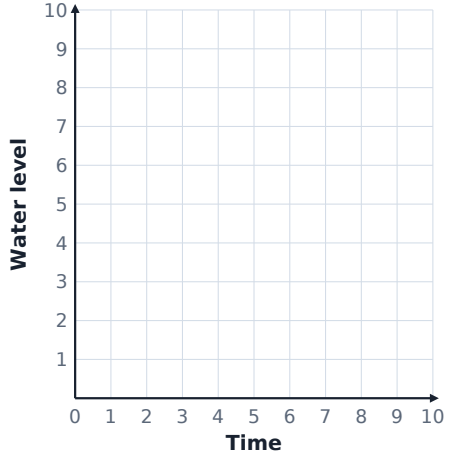
MAIN	Grade 8 main question	**8.F.B.5**
BACKWARD 1	Graph a relationship from a table	**6.EE.C.9**
BACKWARD 2	Plot points that represent a situation	**5.G.A.2**
FORWARD 1	Sketch graphs that show given key features	**HSF-IF.B.4**
FORWARD 2	Graph exponential functions and describe their end behavior	**HSF-IF.C.7.e**

SET 31 • 8.F.B.5

MAIN

Grade 8 main question • Grade 8 • Functions • 8.F.B.5

Sketch a graph of the water level in a bathtub over time: it fills at a steady rate, stays the same while someone bathes, and then drains quickly.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q1/5

938

QUESTION ID

S31-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

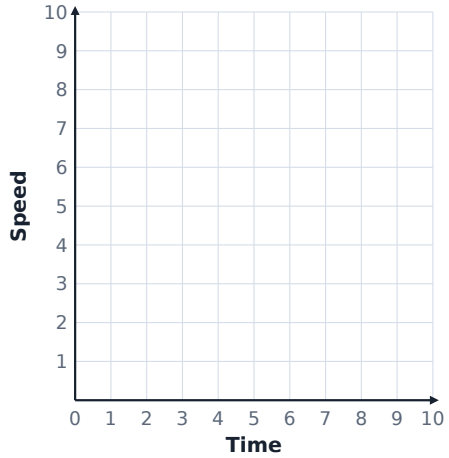
A rising straight segment from 0, then a horizontal segment, then a steeper falling segment back to 0. Any lengths are accepted if the order and steepness are right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q1/5

Sketch a graph of a car's speed over time: it speeds up, travels at a steady speed, and then slows down to a stop.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q2/5

939

QUESTION ID

S31-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

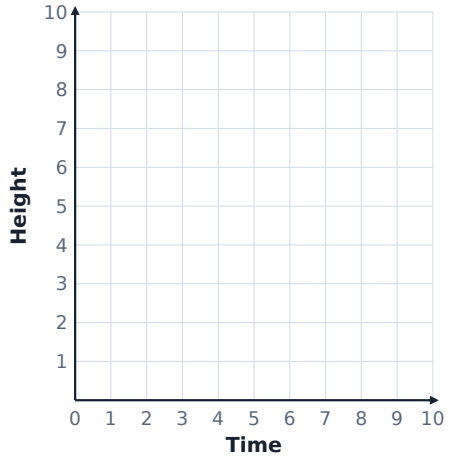
A rising segment from 0, a horizontal segment, then a falling segment back to 0. Any lengths are accepted if the order is right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q2/5

Sketch the height of a ball thrown straight up over time: it rises, slows down, reaches a highest point, and falls back to the ground.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q3/5

940

QUESTION ID

S31-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

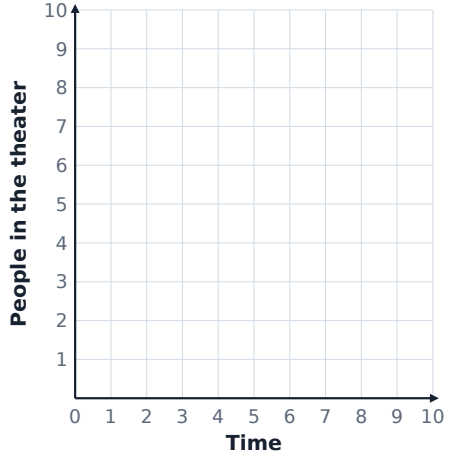
A curve that rises, levels off at one highest point, and falls back to height 0, like an upside-down U. Starting at ground level or at the throwing height is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q3/5

Sketch the number of people in a theater from 6:00 to 10:00. It fills quickly before a 7:00 show, stays full, and empties after the show ends at 9:30.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q4/5

941

QUESTION ID

S31-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A steep rise before 7:00, a horizontal segment from about 7:00 to 9:30, then a steep fall after 9:30 back to about 0. Exact times are not required if the order is right.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

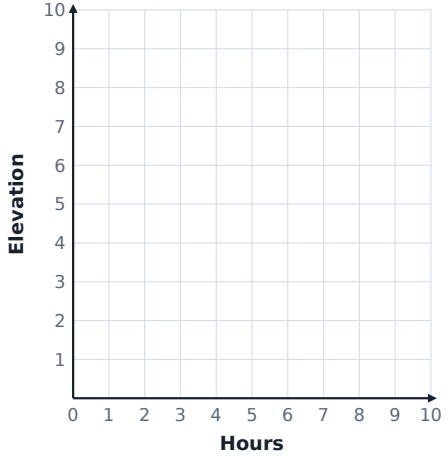
ID S31-M-Q4

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q4/5

Sketch a graph of a hike: the hiker climbs a steep trail for 2 hours, then a gentle slope for 1 hour, then rests for 1 hour.

Is elevation a linear function of time for the whole hike? Explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q5/5

942

QUESTION ID

S31-M-Q5

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: A steeper rising segment from 0 to 2 hours, a less steep rising segment from 2 to 3 hours, and a horizontal segment from 3 to 4 hours

Written: No. The rate of change is different on each part (steep, gentle, then zero), so the whole graph is not one line.

GRADING NOTE Grade both: the three-part graph in the right order and steepness, and "no" with a reason.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-M-Q5

Grade 8 • Functions • 8.F.B.5 • Set 31 • MAIN • Q5/5

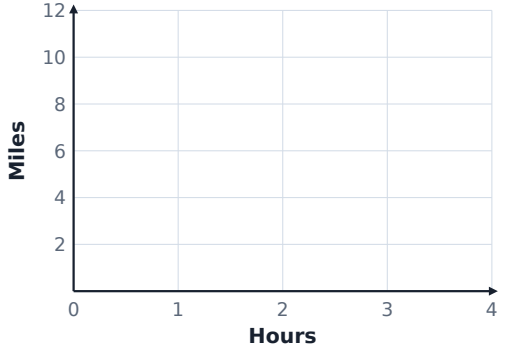
SET 31 • 8.F.B.5

BACKWARD 1

Graph a relationship from a table • Grade 6 • Expressions & Equations • 6.EE.C.9

Graph the pairs in the table.

Hours	0	1	2	3
Miles	0	4	8	12



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q1/5

944

QUESTION ID

S31-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 0), (1, 4), (2, 8), and (3, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q1/5

A hiker walks 4 miles each hour, so miles = 4 × hours.

How many miles does she walk in 2.5 hours?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q2/5

945

QUESTION ID

S31-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q2/5

Which table shows y increasing by the same amount each time?

A. x: 0, 1, 2 and y: 0, 3, 4

B. x: 0, 1, 2 and y: 5, 5, 6

C. x: 0, 1, 2 and y: 6, 3, 9

D. x: 0, 1, 2 and y: 0, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q3/5

946

QUESTION ID

S31-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. x: 0, 1, 2 and y: 0, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q3/5

True or false?

A table where y is 5 for every x shows y not changing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q4/5

947

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q4/5

True or false?

A table with y-values 2, 4, 3 shows y always increasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q5/5

948

QUESTION ID

S31-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 31 • BACKWARD 1 • Q5/5

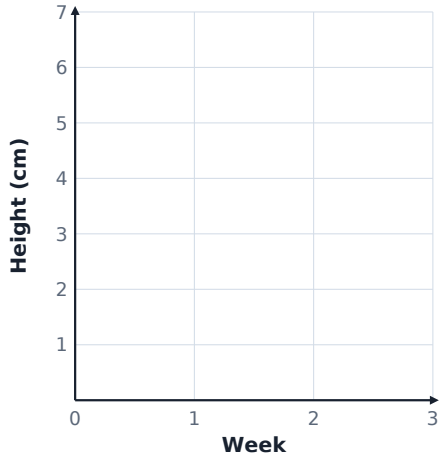
SET 31 • 8.F.B.5

BACKWARD 2

Plot points that represent a situation • Grade 5 • Geometry • 5.G.A.2

A plant is 2 cm tall at week 0, 4 cm at week 1, and 6 cm at week 2.

Plot the points (week, height).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

950

QUESTION ID

S31-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points at (0, 2), (1, 4), and (2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

A plant is 2 cm tall at week 0 and grows 2 cm each week.

Which point shows its height at week 3?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

951

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

On a graph of weeks (x) and plant height in cm (y), what does the point (2, 6) mean?

A. The plant grows 2 cm.

B. At week 6 the plant is 2 cm tall.

C. The plant grows 6 cm each week.

D. At week 2 the plant is 6 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

952

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. At week 2 the plant is 6 cm tall.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

True or false?

On a graph of weeks (x) and height (y), the point (0, 2) shows the height at the start.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

953

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

True or false?

On a graph of weeks (x) and height in cm (y), the point (1, 4) means the plant is 1 cm tall at week 4.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

954

QUESTION ID

S31-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

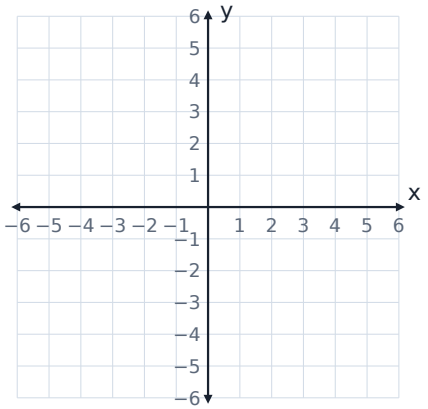
Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 8.F.B.5

FORWARD 1

Sketch graphs that show given key features • High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4

Sketch a function that increases for 0 < x < 3, has a maximum value of 5 at x = 3, and decreases for x > 3.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q1/5

956

QUESTION ID

S31-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

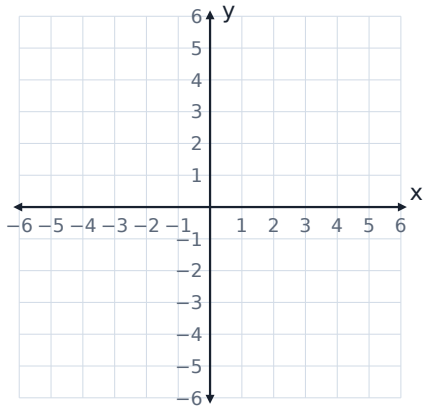
Any graph that rises up to the point (3, 5) and falls after it, with 5 as its highest value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q1/5

Sketch a function whose graph crosses the y-axis at 2 and that increases for all x shown.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q2/5

957

QUESTION ID

S31-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Any always-increasing graph through (0, 2); a line or a curve is accepted

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q2

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q2/5

Which description matches $y = -(x - 2)^2 + 4$?

A. A maximum of 4 at $x = 2$; increasing for $x < 2$

B. A minimum of 4 at $x = 2$

C. A maximum of 2 at $x = 4$

D. Increasing for every x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q3/5

958

QUESTION ID

S31-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A maximum of 4 at x = 2; increasing for x < 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q3

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q3/5

True or false?

A graph with a maximum at x = 3 decreases for some x-values greater than 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q4/5

959

QUESTION ID

S31-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q4

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q4/5

True or false?

A function that is always decreasing can have its maximum in the middle of its domain.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q5/5

960

QUESTION ID

S31-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F1-Q5

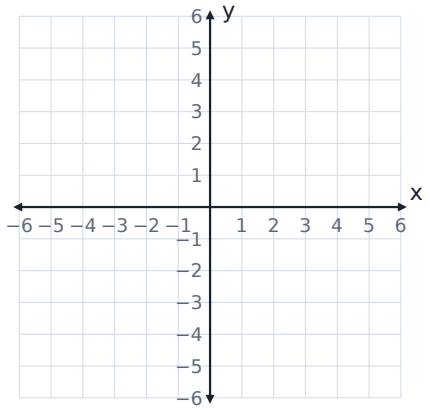
High School (Grade 9 in this bank) • Functions: Interpreting Functions • HSF-IF.B.4 • Set 31 • FORWARD 1 • Q5/5

SET 31 • 8.F.B.5

FORWARD 2

Graph exponential functions and describe their end behavior • High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e

Graph y = 2^x for x from −2 to 2 on the coordinate plane.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q1/5

962

QUESTION ID

S31-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The points (−2, 1/4), (−1, 1/2), (0, 1), (1, 2), and (2, 4) joined by a smooth increasing curve that gets close to the x-axis on the left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q1

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q1/5

What is the y-intercept of the graph of y = 5(0.5)^x?

y-intercept: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q2/5

963

QUESTION ID

S31-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(0, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q2

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q2/5

For y = 3^x, what happens to y as x decreases without bound?

A. y gets closer to 3.

B. y decreases without bound.

C. y increases.

D. y gets closer to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q3/5

964

QUESTION ID

S31-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. y gets closer to 0.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q3

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q3/5

True or false?

The graph of y = 2^x never touches the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q4/5

965

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q4

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q4/5

True or false?

The function y = (0.5)^x is increasing.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q5/5

966

QUESTION ID

S31-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S31-F2-Q5

High School (Grade 10 in this bank) • Functions: Interpreting Functions • HSF-IF.C.7.e • Set 31 • FORWARD 2 • Q5/5

SET 32 • 8.G.A.1.a • GEOMETRY

Verify that rigid motions take lines to lines and segments to segments of the same length

MAIN	Grade 8 main question	**8.G.A.1.a**
BACKWARD 1	Reflect points across the axes	**6.NS.C.6.b**
BACKWARD 2	Find the distance between points with the same x- or y-coordinate	**6.NS.C.8**
FORWARD 1	Define rotations, reflections, and translations precisely	**HSG-CO.A.4**
FORWARD 2	Use rigid motions to decide whether figures are congruent	**HSG-CO.B.6**

SET 32 • 8.G.A.1.a

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.a

Segment AB has endpoints A(1, 2) and B(5, 2). It is translated 3 units down.

What are the endpoints of the image, and how long is the image?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q1/5

969

QUESTION ID

S32-M-Q1

TYPE

Short answer

CORRECT ANSWER

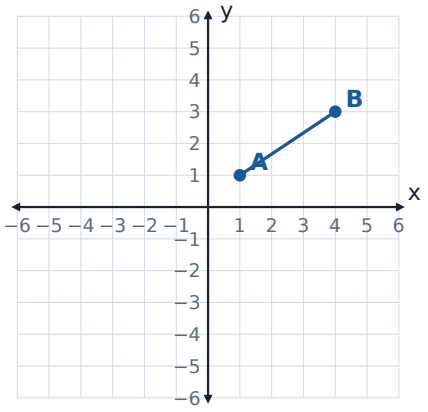
A'(1, −1) and B'(5, −1); it is 4 units long, the same as AB.

GRADING NOTE Grade both the endpoints and the length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q1

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q1/5

Reflect segment AB across the y-axis. Draw the image A'B'.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q2/5

970

QUESTION ID

S32-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

The segment from A'(−1, 1) to B'(−4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q2/5

A segment 7 cm long is rotated 90° about a point.

How long is its image?

A. 3.5 cm

B. 7 cm

C. 14 cm

D. It depends on the center of rotation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q3/5

971

QUESTION ID

S32-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q3/5

True or false?

The image of a straight line under a translation is a straight line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q4/5

QUESTION ID

S32-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q4/5

True or false?

A reflection can map a segment 5 units long to a segment 10 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q5/5

973

QUESTION ID

S32-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Reflections keep lengths the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-M-Q5

Grade 8 • Geometry • 8.G.A.1.a • Set 32 • MAIN • Q5/5

SET 32 • 8.G.A.1.a

BACKWARD 1

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (3, −4) is reflected across the x-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q1/5

975

QUESTION ID

S32-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q1/5

The point (−2, 5) is reflected across the y-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q2/5

976

QUESTION ID

S32-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q2/5

Which point is the reflection of (6, 1) across the y-axis?

A. (1, 6)

B. (6, −1)

C. (−6, −1)

D. (−6, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q3/5

977

QUESTION ID

S32-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−6, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q3/5

True or false?

The reflection of (−3, −3) across the x-axis is (−3, 3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q4/5

978

QUESTION ID

S32-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q4/5

True or false?

The reflection of (4, 2) across the x-axis is (−4, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q5/5

979

QUESTION ID

S32-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 8.G.A.1.a

BACKWARD 2

Find the distance between points with the same x- or y-coordinate • Grade 6 • The Number System • 6.NS.C.8

ANSWER KEY Reduced copy of the question page, with its answer

Find the distance between (1, 2) and (5, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

981

QUESTION ID

S32-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q1/5

Find the distance between (1, −1) and (5, −1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

982

QUESTION ID

S32-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q2/5

Which two points are 6 units apart?

A. (−2, 3) and (2, 3)

B. (0, 1) and (0, 5)

C. (3, −2) and (3, 2)

D. (−2, 3) and (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

983

QUESTION ID

S32-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (−2, 3) and (4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q3/5

True or false?

(0, −3) and (0, 4) are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

984

QUESTION ID

S32-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q4/5

True or false?

(−5, 1) and (2, 1) are 3 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

985

QUESTION ID

S32-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are 7 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-B2-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 8.G.A.1.a

FORWARD 1

Define rotations, reflections, and translations precisely • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4

A reflection across line m maps point P to point P'.

How is line m related to segment PP'?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q1/5

987

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Line m is the perpendicular bisector of PP'.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q1/5

A rotation of 90° about point O maps P to P'.

What is true about OP, OP', and angle POP'?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q2/5

988

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

OP = OP', and angle POP' measures 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q2/5

In a translation, the segments that join each point to its image are...

A. perpendicular to each other

B. all through one point

C. different lengths

D. parallel and the same length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q3/5

989

QUESTION ID

S32-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. parallel and the same length

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q3/5

True or false?

A rotation keeps every point the same distance from the center of rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q4/5

990

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q4/5

True or false?

In a reflection, a point (not on the line) and its image are on the same side of the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q5/5

991

QUESTION ID

S32-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 32 • FORWARD 1 • Q5/5

SET 32 • 8.G.A.1.a

FORWARD 2

Use rigid motions to decide whether figures are congruent • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is mapped onto triangle DEF by a rotation followed by a translation.

What can you conclude about the triangles?

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q1/5

993

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

They are congruent; their corresponding sides and angles are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q1/5

Explain why two figures are congruent when a sequence of rigid motions maps one onto the other.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q2/5

994

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Rigid motions keep all lengths and angle measures, so every part of one figure matches the corresponding part of the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q2/5

Which transformation does NOT always map a figure onto a congruent figure?

A. A translation

B. A rotation

C. A dilation with a scale factor of 2

D. A reflection

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q3/5

995

QUESTION ID

S32-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A dilation with a scale factor of 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q3/5

True or false?

If a translation maps segment AB onto segment CD, then AB = CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q4/5

996

QUESTION ID

S32-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q4/5

True or false?

Two triangles with the same area must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q5/5

997

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S32-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 32 • FORWARD 2 • Q5/5

SET 33 • 8.G.A.1.b • GEOMETRY

Verify that rigid motions take angles to angles of the same measure

MAIN	Grade 8 main question	**8.G.A.1.b**
BACKWARD 1	Find unknown angles with supplementary, complementary, and v…	**7.G.B.5**
BACKWARD 2	Classify triangles by their angles	**4.G.A.2**
FORWARD 1	Specify a sequence of transformations that maps a figure onto a…	**HSG-CO.A.5**
FORWARD 2	Prove theorems about parallelograms	**HSG-CO.C.11**

SET 33 • 8.G.A.1.b

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.b

Angle ABC measures 65°. It is reflected across a line.

What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q1/5

1000

QUESTION ID

S33-M-Q1

TYPE

Short answer

CORRECT ANSWER

65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q1/5

Triangle PQR has angles of 40°, 60°, and 80°. It is rotated 180° about the origin.

What are the angle measures of the image?

Measures: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q2/5

1001

QUESTION ID

S33-M-Q2

TYPE

Short answer

CORRECT ANSWER

40°, 60°, and 80°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q2/5

A right angle is translated 5 units up.

What is its image?

A. A right angle

B. A straight angle

C. An obtuse angle

D. An acute angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q3/5

1002

QUESTION ID

S33-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

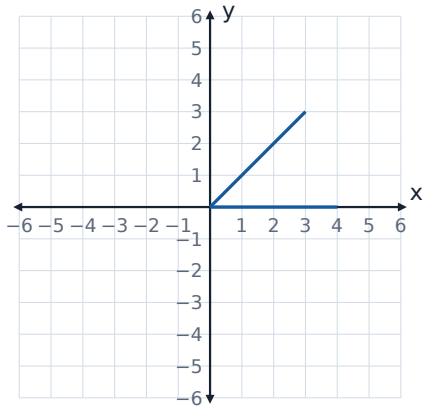
A. A right angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q3/5

The angle has its vertex at the origin. Rotate it 90° counterclockwise about the origin and draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q4/5

1003

QUESTION ID

S33-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

An angle with its vertex at the origin and rays through (0, 4) and (−3, 3); it still measures 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q4/5

True or false?

A rotation can map a 30° angle onto a 60° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q5/5

1004

QUESTION ID

S33-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Rotations keep angle measures the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-M-Q5

Grade 8 • Geometry • 8.G.A.1.b • Set 33 • MAIN • Q5/5

SET 33 • 8.G.A.1.b

BACKWARD 1

Find unknown angles with supplementary, complementary, and vertical angles • Grade 7 • Geometry • 7.G.B.5

Two angles form a straight line. One measures 65°.

What does the other measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q1/5

1006

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

115°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Two lines intersect. One angle measures 40°.

What does the vertical angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q2/5

1007

QUESTION ID

S33-B1-Q2

TYPE

Short answer

CORRECT ANSWER

40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q2/5

What is the complement of a 25° angle?

A. 65°

B. 75°

C. 155°

D. 25°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q3/5

1008

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q3/5

True or false?

Vertical angles have equal measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q4/5

1009

QUESTION ID

S33-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q4/5

True or false?

Supplementary angles add to 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q5/5

1010

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 8.G.A.1.b

BACKWARD 2

Classify triangles by their angles • Grade 4 • Geometry • 4.G.A.2

A triangle has angles of 40°, 60°, and 80°.

Classify it by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q1/5

1012

QUESTION ID

S33-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Acute triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q1/5

A triangle has a 90° angle.

Classify it by its angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q2/5

1013

QUESTION ID

S33-B2-Q2

TYPE

Short answer

CORRECT ANSWER

Right triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q2/5

Which angle measures make an obtuse triangle?

A. 60°, 60°, 60°

B. 30°, 40°, 110°

C. 45°, 45°, 90°

D. 50°, 60°, 70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q3/5

1014

QUESTION ID

S33-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 30°, 40°, 110°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q3/5

True or false?

An acute triangle has three angles that each measure less than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q4/5

1015

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q4/5

True or false?

A right triangle can have an obtuse angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q5/5

1016

QUESTION ID

S33-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 8.G.A.1.b

FORWARD 1

Specify a sequence of transformations that maps a figure onto another • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Triangle A(1, 1), B(3, 1), C(1, 4) is mapped onto A'(−1, −1), B'(−3, −1), C'(−1, −4).

Describe a transformation or sequence of transformations that does this.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q1/5

1018

QUESTION ID

S33-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A 180° rotation about the origin.

GRADING NOTE A reflection across the x-axis followed by a reflection across the y-axis (or the reverse) is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q1/5

Which single transformation maps (2, 3) to (2, −3) and (5, 1) to (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q2/5

1019

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

A reflection across the x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Which rotation about the origin maps (4, 0) to (0, 4)?

A. 270° counterclockwise

B. 180°

C. 90° clockwise

D. 90° counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q3/5

1020

QUESTION ID

S33-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 90° counterclockwise

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across a line and then reflecting the image across the same line gives back the original figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q4/5

1021

QUESTION ID

S33-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

A translation 3 units right followed by a translation 3 units left moves the figure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q5/5

1022

QUESTION ID

S33-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The figure returns to where it started.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.5 • Set 33 • FORWARD 1 • Q5/5

SET 33 • 8.G.A.1.b

FORWARD 2

Prove theorems about parallelograms • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11

ANSWER KEY Reduced copy of the question page, with its answer

ABCD is a parallelogram. Prove that angle A and angle B are supplementary. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q1/5

1024

QUESTION ID

S33-F2-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle A + angle B = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: AD ∥ BC (definition of a parallelogram), and AB is a transversal; angles A and B are same-side interior angles, so they are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In parallelogram ABCD, angle A = 70°.

Find the other three angles and give a reason for each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q2/5

1025

QUESTION ID

S33-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Angle B = 110° and angle D = 110° (consecutive angles are supplementary); angle C = 70° (opposite angles are congruent).

GRADING NOTE Grade the angles and the reasons.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q2/5

Which statement about every parallelogram can be proved?

A. All of its angles are right angles.

B. Its diagonals are congruent.

C. Its diagonals bisect each other.

D. Its diagonals are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q3/5

1026

QUESTION ID

S33-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Its diagonals bisect each other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q3/5

True or false?

Opposite sides of a parallelogram are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q4/5

1027

QUESTION ID

S33-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q4/5

True or false?

A quadrilateral with one pair of parallel sides must be a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q5/5

1028

QUESTION ID

S33-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S33-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.C.11 • Set 33 • FORWARD 2 • Q5/5

SET 34 • 8.G.A.1.c • GEOMETRY

Verify that rigid motions take parallel lines to parallel lines

MAIN	Grade 8 main question	**8.G.A.1.c**
BACKWARD 1	Identify parallel lines and segments	**4.G.A.1**
BACKWARD 2	Draw polygons in the coordinate plane and find side lengths	**6.G.A.3**
FORWARD 1	Prove theorems about angles formed by parallel lines	**HSG-CO.C.9**
FORWARD 2	Use coordinates to prove simple geometric theorems	**HSG-GPE.B.4**

SET 34 • 8.G.A.1.c

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.1.c

Lines p and q are parallel. Both are rotated 90° about the same point.

Are the images parallel? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q1/5

1031

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

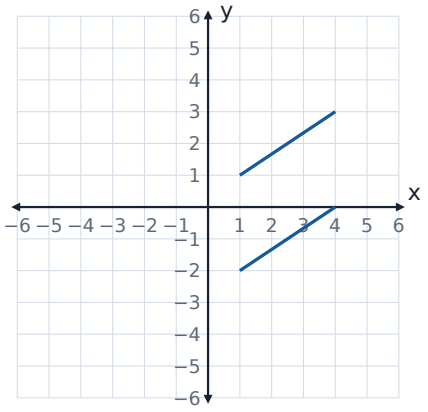
Yes. Rigid motions take parallel lines to parallel lines; both lines turn by the same angle, so the images never meet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q1/5

The two segments are parallel. Translate both segments 4 units to the left and draw the images.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q2/5

1032

QUESTION ID

S34-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Segments from (−3, 1) to (0, 3) and from (−3, −2) to (0, 0); the images are still parallel

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q2/5

A parallelogram is reflected across a line.

What is its image?

A. A rectangle, in every case

B. A figure with no parallel sides

C. A parallelogram congruent to the original

D. A trapezoid with only one pair of parallel sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q3/5

1033

QUESTION ID

S34-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A parallelogram congruent to the original

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q3

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q3/5

The line y = 2x + 1 is translated 3 units up.

What is the equation of the image? Is the image parallel to the original line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q4/5

1034

QUESTION ID

S34-M-Q4

TYPE

Short answer

CORRECT ANSWER

y = 2x + 4; yes, the slope is still 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-M-Q4

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q4/5

True or false?

A rigid motion can map two parallel lines onto two lines that intersect.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q5/5

1035

QUESTION ID

S34-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 8 • Geometry • 8.G.A.1.c • Set 34 • MAIN • Q5/5

SET 34 • 8.G.A.1.c

BACKWARD 1

Identify parallel lines and segments • Grade 4 • Geometry • 4.G.A.1

Name a quadrilateral that has two pairs of parallel sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q1/5

1037

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle

GRADING NOTE Any parallelogram (square, rhombus, rectangle) is correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q1/5

How many pairs of parallel sides does a trapezoid that is not a parallelogram have?

Pairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q2/5

1038

QUESTION ID

S34-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q2/5

Which capital letter contains a pair of parallel segments?

A. V

B. H

C. T

D. L

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q3/5

1039

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. H

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q3/5

True or false?

Parallel lines never meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q4/5

1040

QUESTION ID

S34-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q4/5

True or false?

Perpendicular lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q5/5

1041

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

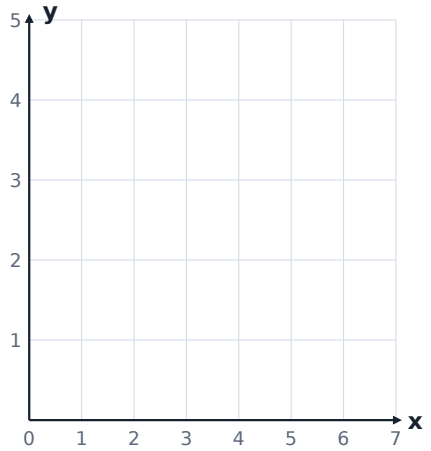
Grade 4 • Geometry • 4.G.A.1 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 8.G.A.1.c

BACKWARD 2

Draw polygons in the coordinate plane and find side lengths • Grade 6 • Geometry • 6.G.A.3

Draw the quadrilateral with vertices (1, 1), (5, 1), (6, 4), and (2, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q1/5

1043

QUESTION ID

S34-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The quadrilateral with those four vertices (a parallelogram)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q1/5

A quadrilateral has vertices (1, 1), (5, 1), (6, 4), and (2, 4).

How long is the side from (1, 1) to (5, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q2/5

1044

QUESTION ID

S34-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q2/5

A quadrilateral has vertices (1, 1), (5, 1), (6, 4), and (2, 4).

Which side is parallel to the side from (1, 1) to (5, 1)?

A. From (1, 1) to (6, 4)

B. From (2, 4) to (6, 4)

C. From (1, 1) to (2, 4)

D. From (5, 1) to (6, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q3/5

1045

QUESTION ID

S34-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. From (2, 4) to (6, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q3/5

True or false?

The side from (2, 4) to (6, 4) is 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q4/5

1046

QUESTION ID

S34-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q4/5

True or false?

The segment from (1, 1) to (1, 4) is 4 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q5/5

1047

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 3 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-B2-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 34 • BACKWARD 2 • Q5/5

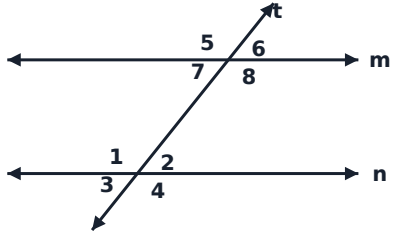
SET 34 • 8.G.A.1.c

FORWARD 1

Prove theorems about angles formed by parallel lines • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 2 = angle 7. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q1/5

1049

QUESTION ID

S34-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 = angle 7

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 2 = angle 6 (corresponding angles, m ∥ n); angle 6 = angle 7 (vertical angles); so angle 2 = angle 7.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q1/5

Two parallel lines are cut by a transversal. One angle measures 72°.

What does the same-side interior angle measure? Give the reason.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q2/5

1050

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

108°; same-side interior angles are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q2/5

Which statement can be proved from "corresponding angles formed by parallel lines are congruent" and "vertical angles are congruent"?

A. Alternate interior angles are congruent.

B. The transversal is parallel to the lines.

C. The lines are perpendicular.

D. All the angles are right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q3/5

1051

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Alternate interior angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q3/5

True or false?

If alternate interior angles formed by a transversal are congruent, then the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q4/5

1052

QUESTION ID

S34-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q4/5

True or false?

Same-side interior angles formed by parallel lines are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q5/5

1053

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE They are supplementary.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 34 • FORWARD 1 • Q5/5

SET 34 • 8.G.A.1.c

FORWARD 2

Use coordinates to prove simple geometric theorems • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Show that the quadrilateral with vertices (0, 0), (4, 0), (5, 3), and (1, 3) is a parallelogram.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q1/5

1055

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The sides from (0, 0) to (4, 0) and from (1, 3) to (5, 3) both have slope 0; the sides from (0, 0) to (1, 3) and from (4, 0) to (5, 3) both have slope 3. Both pairs of opposite sides are parallel.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q1/5

Is the triangle with vertices (0, 0), (4, 0), and (0, 3) a right triangle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q2/5

1056

QUESTION ID

S34-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. One side is horizontal and one is vertical, so they are perpendicular.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q2/5

In order, (0, 0), (2, 0), (3, 2), and a fourth point form a parallelogram.

Which is the fourth point?

A. (0, 3)

B. (3, 0)

C. (1, 2)

D. (2, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q3/5

1057

QUESTION ID

S34-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S34-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q3/5

True or false?

A quadrilateral whose opposite sides have equal slopes is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q4/5

1058

QUESTION ID

S34-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q4/5

True or false?

The points (0, 0), (1, 1), and (2, 3) lie on one line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q5/5

1059

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S34-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 34 • FORWARD 2 • Q5/5

SET 35 • 8.G.A.2 • GEOMETRY

Describe a sequence of rigid motions that shows two figures are congruent

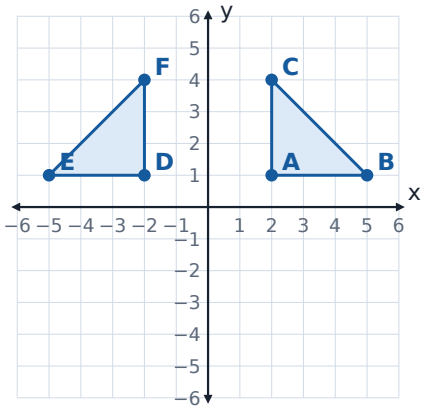
MAIN	Grade 8 main question	**8.G.A.2**
BACKWARD 1	Draw a polygon from its vertices	**6.G.A.3**
BACKWARD 2	Use signs to locate points across the axes	**6.NS.C.6.b**
FORWARD 1	Use corresponding parts to explain triangle congruence	**HSG-CO.B.7**
FORWARD 2	Use triangle congruence criteria	**HSG-CO.B.8**

SET 35 • 8.G.A.2

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.2

Describe a rigid motion that maps triangle ABC onto triangle DEF.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q1/5

1062

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

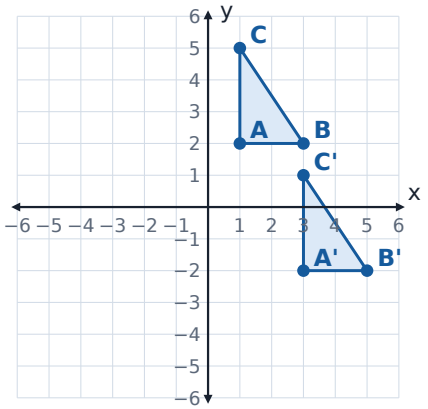
A reflection across the y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q1/5

Describe a rigid motion that maps triangle ABC onto triangle A'B'C'.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q2/5

1063

QUESTION ID

S35-M-Q2

TYPE

Short answer

CORRECT ANSWER

A translation 2 units right and 4 units down

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q2/5

Figure P is mapped onto figure Q by a 90° rotation followed by a translation.

Which statement is true?

A. P and Q are congruent.

B. Q is larger than P.

C. P and Q are not congruent because they face different directions.

D. P and Q have different angle measures.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q3/5

1064

QUESTION ID

S35-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. P and Q are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q3/5

Explain why a triangle and its image after a dilation with a scale factor of 2 are not congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q4/5

1065

QUESTION ID

S35-M-Q4

TYPE

Short answer

CORRECT ANSWER

The dilation doubles every side length, but rigid motions (which show congruence) keep lengths the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-M-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q4/5

1065

True or false?

Two figures are congruent if a sequence of rotations, reflections, and translations maps one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q5/5

1066

QUESTION ID

S35-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

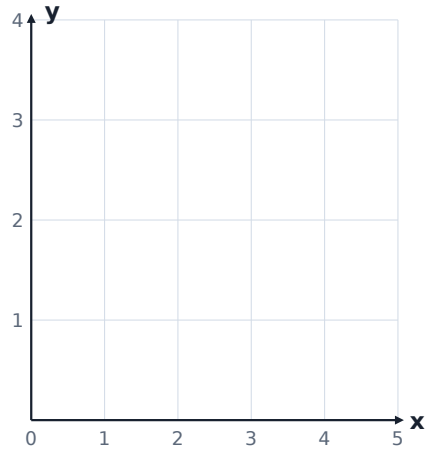
Grade 8 • Geometry • 8.G.A.2 • Set 35 • MAIN • Q5/5

SET 35 • 8.G.A.2

BACKWARD 1

Draw a polygon from its vertices • Grade 6 • Geometry • 6.G.A.3

Draw the triangle with vertices (1, 1), (4, 1), and (1, 3).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q1/5

1068

QUESTION ID

S35-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with those three vertices

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q1/5

A triangle has vertices (1, 1), (4, 1), and (1, 3).

How long is the side from (1, 1) to (4, 1)?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q2/5

1069

QUESTION ID

S35-B1-Q2

TYPE

Short answer

CORRECT ANSWER

3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q2/5

A rectangle has vertices (0, 0), (5, 0), and (5, 2).

What is the fourth vertex?

A. (0, 2)

B. (2, 0)

C. (5, 5)

D. (2, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q3/5

1070

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q3/5

True or false?

The side from (1, 1) to (1, 3) is 2 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q4/5

1071

QUESTION ID

S35-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q4/5

True or false?

The points (2, 1), (2, 5), and (6, 1) form a triangle with a side 3 units long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q5/5

1072

QUESTION ID

S35-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its sides along the grid lines are 4 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 8.G.A.2

BACKWARD 2

Use signs to locate points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (5, 2) is reflected across the y-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q1/5

1074

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−5, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q1/5

The point $(-1, -4)$ is reflected across the x-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q2/5

1075

QUESTION ID

S35-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$(-1, 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q2/5

Points (3, 2) and (3, −2) differ only in the sign of the y-coordinate.

How are they related?

A. They are reflections across the x-axis.

B. They are reflections across the y-axis.

C. They are the same point.

D. They are 3 units apart.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q3/5

1076

QUESTION ID

S35-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. They are reflections across the x-axis.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q3/5

True or false?

(−4, 1) and (4, 1) are reflections of each other across the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q4/5

1077

QUESTION ID

S35-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q4/5

True or false?

(2, −6) and (−2, 6) are reflections of each other across the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q5/5

1078

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 8.G.A.2

FORWARD 1

Use corresponding parts to explain triangle congruence • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7

Triangle ABC is congruent to triangle DEF, with AB = 5 and angle B = 40°.

Which side and which angle of DEF must equal these?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q1/5

1080

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

DE = 5 and angle E = 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q1/5

Two triangles have all corresponding sides and all corresponding angles congruent.

What can you conclude? Explain with rigid motions.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q2/5

1081

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The triangles are congruent; a sequence of rigid motions maps one onto the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q2/5

Triangle PQR is congruent to triangle XYZ.

Which statement must be true?

A. PQ = YZ

B. QR = YZ

C. Angle P = angle Z

D. PR = XY

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q3/5

1082

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. QR = YZ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q3/5

True or false?

Corresponding parts of congruent triangles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q4/5

1083

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q4/5

True or false?

If triangle ABC is congruent to triangle DEF, then AC = EF.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q5/5

1084

QUESTION ID

S35-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE AC = DF.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.B.7 • Set 35 • FORWARD 1 • Q5/5

SET 35 • 8.G.A.2

FORWARD 2

Use triangle congruence criteria • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8

Two triangles have AB = DE, angle A = angle D, and AC = DF.

Which criterion shows that they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q1/5

1086

QUESTION ID

S35-F2-Q1

TYPE

Short answer

CORRECT ANSWER

SAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q1/5

Two triangles have angle A = angle D, AB = DE, and angle B = angle E.

Which criterion shows that they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q2/5

1087

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

ASA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q2/5

Which is NOT a valid congruence criterion for triangles?

A. AAA

B. ASA

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q3/5

1088

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. AAA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q3/5

True or false?

SSS means that three pairs of equal sides show two triangles are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q4/5

1089

QUESTION ID

S35-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q4/5

True or false?

Two sides and an angle that is not between them (SSA) always prove two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q5/5

1090

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S35-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.8 • Set 35 • FORWARD 2 • Q5/5

SET 36 • 8.G.A.3 • GEOMETRY

Translate figures on the coordinate plane

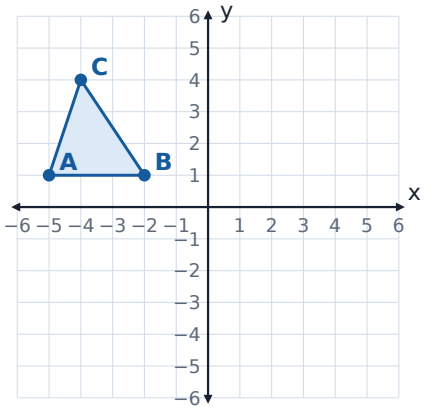
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Find and plot points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Add integers	**7.NS.A.1.b**
FORWARD 1	Describe transformations as functions and decide which preserve…	**HSG-CO.A.2**
FORWARD 2	Find the point that divides a segment in a given ratio	**HSG-GPE.B.6**

SET 36 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Translate triangle ABC 4 units right and 2 units down. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q1/5

1093

QUESTION ID

S36-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(−1, −1), B'(2, −1), and C'(0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q1/5

The point P(3, −2) is translated 5 units left and 4 units up.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q2/5

1094

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−2, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q2/5

Which rule describes a translation 6 units down?

A. $(x, y) \to (x, y - 6)$

B. $(x, y) \to (x, y + 6)$

C. $(x, y) \to (x + 6, y)$

D. $(x, y) \to (x - 6, y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q3/5

1095

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $(x, y) \to (x, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q3/5

A translation maps A(2, 5) to A'(−1, 7).

Where does it map B(4, 0)?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q4/5

1096

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

(1, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q4/5

True or false?

The rule (x, y) → (x + 3, y − 1) moves every point 3 units right and 1 unit down.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q5/5

1097

QUESTION ID

S36-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

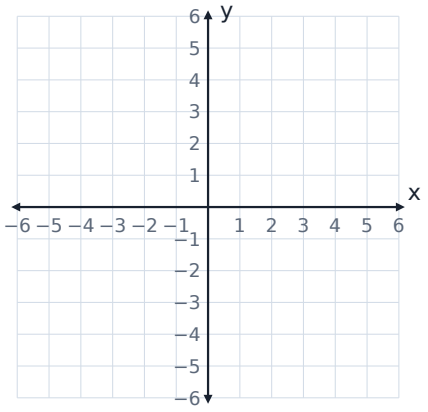
Grade 8 • Geometry • 8.G.A.3 • Set 36 • MAIN • Q5/5

SET 36 • 8.G.A.3

BACKWARD 1

Find and plot points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot the points (−3, 2) and (4, −5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q1/5

1099

QUESTION ID

S36-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

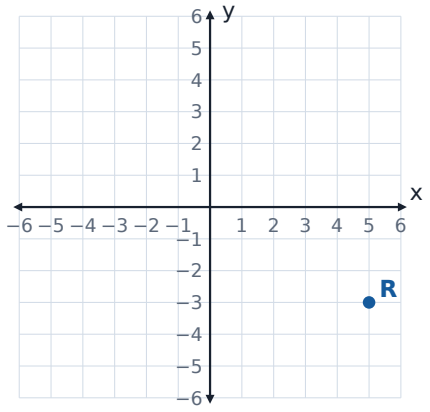
Points at (−3, 2) and (4, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q1/5

What are the coordinates of point R?



Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q2/5

1100

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(5, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q2/5

Which point is 2 units to the left of (1, 4)?

A. (1, 6)

B. (−1, 4)

C. (1, 2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q3/5

1101

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (−1, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q3/5

True or false?

Moving 3 units up from (2, −1) gives (2, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q4/5

1102

QUESTION ID

S36-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q4/5

True or false?

Moving 4 units left from (0, 0) gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q5/5

1103

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 8.G.A.3

BACKWARD 2

Add integers • Grade 7 • The Number System • 7.NS.A.1.b

ANSWER KEY Reduced copy of the question page, with its answer

Add.

$-4 + 4$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q1/5

1105

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q1/5

Add.

$3 + (-5)$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q2/5

1106

QUESTION ID

S36-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q2/5

Add.

$-1 + 6$

A. 5

B. -5

C. 7

D. -7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q3/5

1107

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q3/5

True or false?

$-2 + (-3) = -5$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q4/5

1108

QUESTION ID

S36-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q4/5

True or false?

$5 + (-8) = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q5/5

1109

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 8.G.A.3

FORWARD 1

Describe transformations as functions and decide which preserve distance • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2

Write the rule for a translation 2 units left and 7 units up.

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q1/5

1111

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(x, y) \rightarrow (x - 2, y + 7)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q1/5

Does the rule (x, y) → (2x, y) keep distances the same? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q2/5

1112

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No. It doubles horizontal distances; the segment from (0, 0) to (1, 0) becomes 2 units long.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q2/5

Which transformation keeps all distances and angles the same?

A. $(x, y) \to (x, 2y)$

B. $(x, y) \to (x^2, y)$

C. $(x, y) \to (3x, 3y)$

D. $(x, y) \to (x + 4, y - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q3/5

1113

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (x + 4, y − 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q3/5

True or false?

A translation is a function whose inputs and outputs are points of the plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q4/5

1114

QUESTION ID

S36-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q4/5

True or false?

The rule (x, y) → (x + 1, y + 1) changes the length of every segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q5/5

1115

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.2 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 8.G.A.3

FORWARD 2

Find the point that divides a segment in a given ratio • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6

Find the midpoint of the segment from (2, 4) to (8, 10).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q1/5

1117

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q1/5

Find the point that is $\frac{1}{3}$ of the way from A(0, 0) to B(9, 6).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q2/5

1118

QUESTION ID

S36-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q2/5

Point P is on segment AB from A(1, 1) to B(7, 4), with AP : PB = 2 : 1.

What are the coordinates of P?

A. (3, 2)

B. (4, 2.5)

C. (6, 3)

D. (5, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q3/5

1119

QUESTION ID

S36-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (5, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S36-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q3/5

True or false?

The midpoint of the segment from (−4, 2) to (6, −2) is (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q4/5

1120

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q4/5

True or false?

The point halfway from (0, 0) to (10, 4) is (10, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q5/5

1121

QUESTION ID

S36-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The midpoint is (5, 2).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S36-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.6 • Set 36 • FORWARD 2 • Q5/5

SET 37 • 8.G.A.3 • GEOMETRY

Reflect figures across the x-axis and the y-axis

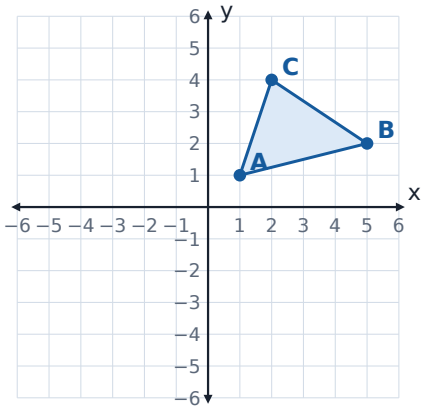
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Locate points that are reflections across the axes	**6.NS.C.6.b**
BACKWARD 2	Find the opposite of a number	**6.NS.C.6.a**
FORWARD 1	Describe the rotations and reflections that carry a polygon onto it…	**HSG-CO.A.3**
FORWARD 2	Draw the image of a figure after a sequence of transformations	**HSG-CO.B.6**

SET 37 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Reflect triangle ABC across the x-axis. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q1/5

1124

QUESTION ID

S37-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(1, −1), B'(5, −2), and C'(2, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q1/5

The point $(-3, 5)$ is reflected across the y-axis.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q2/5

1125

QUESTION ID

S37-M-Q2

TYPE

Short answer

CORRECT ANSWER

(3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q2/5

Which rule describes a reflection across the x-axis?

A. (x, y) → (−x, y)

B. (x, y) → (−x, −y)

C. (x, y) → (y, x)

D. (x, y) → (x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q3/5

1126

QUESTION ID

S37-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q3/5

A segment has endpoints (2, 3) and (5, 1). It is reflected across the y-axis.

What are the endpoints of the image?

Endpoints: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q4/5

1127

QUESTION ID

S37-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−2, 3) and (−5, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q4/5

True or false?

Reflecting (4, −6) across the x-axis gives (−4, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q5/5

1128

QUESTION ID
S37-M-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE The image is (4, 6).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 37 • MAIN • Q5/5

SET 37 • 8.G.A.3

BACKWARD 1

Locate points that are reflections across the axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (7, −2) is reflected across the x-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q1/5

1130

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(7, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q1/5

The point (−6, −3) is reflected across the y-axis.

What is its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q2/5

1131

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

(6, −3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q2/5

Reflecting a point across the y-axis changes...

A. the sign of the y-coordinate

B. neither sign

C. the sign of the x-coordinate

D. both signs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q3/5

1132

QUESTION ID

S37-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. the sign of the x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q3/5

True or false?

(0, 5) is its own reflection across the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q4/5

1133

QUESTION ID

S37-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q4/5

True or false?

The reflection of (−1, 8) across the y-axis is (−1, −8).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q5/5

1134

QUESTION ID

S37-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 8.G.A.3

BACKWARD 2

Find the opposite of a number • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of −7?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q1/5

1136

QUESTION ID

S37-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q1/5

What is the opposite of 2.5?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q2/5

1137

QUESTION ID

S37-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q2/5

Which pair are opposites?

A. −4 and 4

B. 4 and $\frac{1}{4}$

C. 0 and 1

D. −4 and −4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q3/5

1138

QUESTION ID

S37-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. −4 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q3/5

True or false?

The opposite of the opposite of 3 is 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q4/5

1139

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q4/5

True or false?

The opposite of 0 is −1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q5/5

1140

QUESTION ID

S37-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 8.G.A.3

FORWARD 1

Describe the rotations and reflections that carry a polygon onto itself • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3

How many lines of symmetry does a square have?

Lines: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q1/5

1142

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q1/5

Which rotations (between 0° and 360°) about its center carry a regular hexagon onto itself?

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q2/5

1143

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

60°, 120°, 180°, 240°, and 300°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q2/5

Which figure is carried onto itself by a 180° rotation but has no line of symmetry?

A. A parallelogram that is not a rectangle or a rhombus

B. A square

C. An equilateral triangle

D. An isosceles trapezoid

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q3/5

1144

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A parallelogram that is not a rectangle or a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q3/5

True or false?

A rectangle that is not a square has exactly 2 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q4/5

1145

QUESTION ID

S37-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

An isosceles trapezoid is carried onto itself by a 90° rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q5/5

1146

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

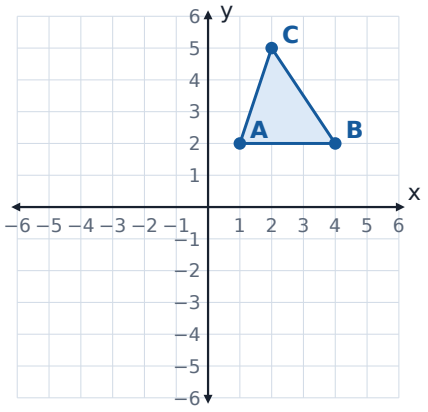
High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 8.G.A.3

FORWARD 2

Draw the image of a figure after a sequence of transformations • High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6

Reflect triangle ABC across the x-axis, then reflect that image across the y-axis. Draw the final image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q1/5

1148

QUESTION ID

S37-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A''(−1, −2), B''(−4, −2), and C''(−2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q1

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q1/5

What single transformation has the same effect as a reflection across the x-axis followed by a reflection across the y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q2/5

1149

QUESTION ID

S37-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q2

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q2/5

The point (3, 1) is reflected across the line y = x.

What is its image?

A. (3, −1)

B. (−1, −3)

C. (−3, 1)

D. (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q3/5

1150

QUESTION ID

S37-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q3

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q3/5

True or false?

Reflecting a figure across one line and then across a parallel line has the same effect as a translation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q4/5

1151

QUESTION ID

S37-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q4/5

True or false?

Doing a translation and a reflection in the other order always gives the same image.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q5/5

1152

QUESTION ID

S37-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S37-F2-Q5

High School (Grade 10 in this bank) • Geometry: Congruence • HSG-CO.B.6 • Set 37 • FORWARD 2 • Q5/5

SET 38 • 8.G.A.3 • GEOMETRY

Rotate figures 90° and 180° about the origin

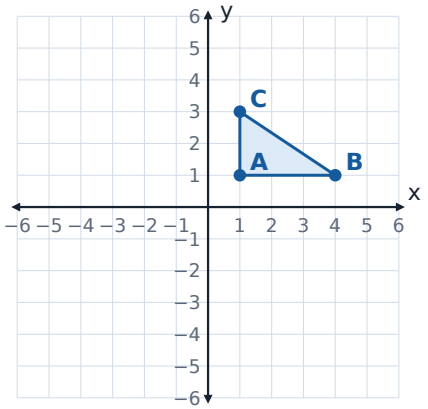
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Relate turns to angle measures	**4.MD.C.5.a**
BACKWARD 2	Locate a point reflected across both axes	**6.NS.C.6.b**
FORWARD 1	Describe a rotation in terms of distance and angle	**HSG-CO.A.4**
FORWARD 2	Use relationships among central angles, inscribed angles, and rad…	**HSG-C.A.2**

SET 38 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Rotate triangle ABC 90° counterclockwise about the origin. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q1/5

1155

QUESTION ID

S38-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(−1, 1), B'(−1, 4), and C'(−3, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q1/5

The point (5, −2) is rotated 180° about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q2/5

1156

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

(−5, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q2/5

Which rule describes a 90° clockwise rotation about the origin?

A. (x, y) → (y, x)

B. (x, y) → (y, −x)

C. (x, y) → (−y, x)

D. (x, y) → (−x, −y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q3/5

1157

QUESTION ID

S38-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. (x, y) → (y, −x)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q3/5

The point (2, 3) is rotated 90° counterclockwise about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q4/5

1158

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

(−3, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q4/5

True or false?

Rotating $(-4, 1)$ by $180°$ about the origin gives $(4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q5/5

1159

QUESTION ID

S38-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 38 • MAIN • Q5/5

SET 38 • 8.G.A.3

BACKWARD 1

Relate turns to angle measures • Grade 4 • Measurement & Data • 4.MD.C.5.a

How many degrees are in a quarter turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q1/5

1161

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q1/5

How many degrees are in a half turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q2/5

1162

QUESTION ID

S38-B1-Q2

TYPE

Short answer

CORRECT ANSWER

180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q2/5

How many degrees are in a three-quarter turn?

A. 360°

B. 180°

C. 270°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q3/5

1163

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q3/5

True or false?

A full turn is 360°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q4/5

1164

QUESTION ID

S38-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q4/5

True or false?

A half turn is 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q5/5

1165

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 38 • BACKWARD 1 • Q5/5

SET 38 • 8.G.A.3

BACKWARD 2

Locate a point reflected across both axes • Grade 6 • The Number System • 6.NS.C.6.b

The point (3, 4) is reflected across the x-axis and then across the y-axis.

What is the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q1/5

1167

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(−3, −4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q1/5

The point (−2, 5) is reflected across both axes.

What is the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q2/5

1168

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q2/5

Reflecting a point across both axes changes...

A. the signs of both coordinates

B. only the y-coordinate

C. neither coordinate

D. only the x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q3/5

1169

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. the signs of both coordinates

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q3/5

True or false?

Reflecting (1, −6) across both axes gives (−1, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q4/5

1170

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q4/5

True or false?

Reflecting (4, 4) across both axes gives (4, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q5/5

1171

QUESTION ID

S38-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 8.G.A.3

FORWARD 1

Describe a rotation in terms of distance and angle • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4

A rotation of 120° about point O maps A to A'.

Which lengths and which angle are determined by the rotation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q1/5

1173

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

OA = OA', and angle AOA' = 120°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q1/5

P(6, 0) is rotated 60° counterclockwise about the origin.

How far is the image from the origin?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q2/5

1174

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q2/5

A rotation about point O maps every point P to a point P' so that...

A. P' is on the line of reflection

B. P' is twice as far from O as P

C. OP = OP' and angle POP' is the angle of rotation

D. PP' is parallel to one fixed line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q3/5

1175

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. OP = OP' and angle POP' is the angle of rotation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q3/5

True or false?

In a rotation by a nonzero angle less than 360°, the center is the only point that does not move.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q4/5

1176

QUESTION ID

S38-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q4/5

True or false?

A 90° rotation maps every horizontal segment to another horizontal segment.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q5/5

1177

QUESTION ID
S38-F1-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE It maps horizontal segments to vertical segments.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.A.4 • Set 38 • FORWARD 1 • Q5/5

SET 38 • 8.G.A.3

FORWARD 2

Use relationships among central angles, inscribed angles, and radii • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2

A central angle of a circle measures 80°.

What does an inscribed angle that intercepts the same arc measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q1/5

1179

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q1/5

An inscribed angle intercepts a semicircle.

What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q2/5

1180

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q2/5

A radius is drawn to the point where a tangent line touches a circle.

What angle do the radius and the tangent line form?

A. 90°

B. 45°

C. 180°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q3/5

1181

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q3/5

True or false?

Rotating a circle about its center by any angle carries the circle onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q4/5

1182

QUESTION ID

S38-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q4/5

True or false?

An inscribed angle measures twice its intercepted arc.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q5/5

1183

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It measures half of its intercepted arc.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S38-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.2 • Set 38 • FORWARD 2 • Q5/5

SET 39 • 8.G.A.3 • GEOMETRY

Dilate figures on the coordinate plane with the origin as the center

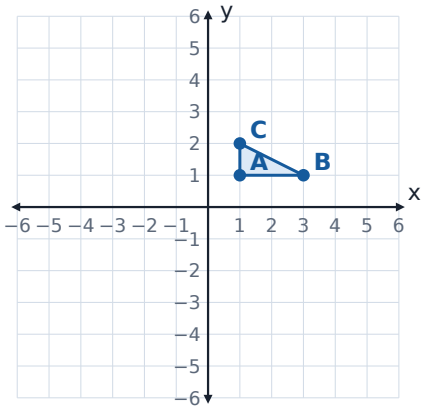
MAIN	Grade 8 main question	**8.G.A.3**
BACKWARD 1	Use a scale factor to find lengths in a scale drawing	**7.G.A.1**
BACKWARD 2	Multiply rational numbers	**7.NS.A.2.a**
FORWARD 1	Verify the properties of dilations	**HSG-SRT.A.1**
FORWARD 2	Use similar triangles to solve problems	**HSG-SRT.B.5**

SET 39 • 8.G.A.3

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.3

Dilate triangle ABC by a scale factor of 2 with the origin as the center. Draw the image.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q1/5

1186

QUESTION ID

S39-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

The triangle with vertices A'(2, 2), B'(6, 2), and C'(2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q1/5

The point (6, −4) is dilated by a scale factor of $\frac{1}{2}$ with the origin as the center.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q2/5

1187

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

(3, −2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q2/5

Which rule describes a dilation with a scale factor of 3 and the origin as the center?

A. (x, y) → (3x, y)

B. (x, y) → (x + 3, y + 3)

C. (x, y) → $(\frac{x}{3}, \frac{y}{3})$

D. (x, y) → (3x, 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q3/5

1188

QUESTION ID

S39-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (x, y) → (3x, 3y)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q3/5

A dilation with the origin as the center maps (4, 2) to (10, 5).

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q4/5

1189

QUESTION ID

S39-M-Q4

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE 5/2 is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-M-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q4/5

True or false?

A dilation with a scale factor of $\frac{1}{3}$ and the origin as the center maps (9, 6) to (3, 2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q5/5

1190

QUESTION ID

S39-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 39 • MAIN • Q5/5

SET 39 • 8.G.A.3

BACKWARD 1

Use a scale factor to find lengths in a scale drawing • Grade 7 • Geometry • 7.G.A.1

A 4 cm segment is enlarged by a scale factor of 3.

How long is the new segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q1/5

1192

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q1/5

A 10 cm segment is reduced by a scale factor of $\frac{1}{2}$.

How long is the new segment?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q2/5

1193

QUESTION ID

S39-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q2/5

A rectangle 2 cm by 5 cm is enlarged by a scale factor of 4.

What are the new dimensions?

A. 8 cm by 20 cm

B. 8 cm by 5 cm

C. 6 cm by 9 cm

D. 2 cm by 20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q3/5

1194

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 cm by 20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q3/5

True or false?

A scale factor greater than 1 makes a figure larger.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q4/5

1195

QUESTION ID

S39-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q4/5

True or false?

A scale factor of $\frac{1}{4}$ makes each length 4 times as long.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q5/5

1196

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 8.G.A.3

BACKWARD 2

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.a

Multiply.

$2 \times (-4)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q1/5

1198

QUESTION ID

S39-B2-Q1

TYPE

Short answer

CORRECT ANSWER

−8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q1/5

Multiply.

$\frac{1}{2} \times (-6)$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q2/5

1199

QUESTION ID

S39-B2-Q2

TYPE

Short answer

CORRECT ANSWER

−3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q2/5

Multiply.

3.5×4

A. 12

B. 0.875

C. 14

D. 7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q3/5

1200

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-B2-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q3/5

True or false?

$3 \times (-2) = -6$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q4/5

1201

QUESTION ID

S39-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3} \times 9 = 27$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q5/5

1202

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 39 • BACKWARD 2 • Q5/5

SET 39 • 8.G.A.3

FORWARD 1

Verify the properties of dilations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1

A dilation with the origin as the center and a scale factor of 3 is applied to the line y = x + 1.

What is the equation of the image?

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q1/5

1204

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

y = x + 3 (a line parallel to the original)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q1**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q1/5

A segment 4 units long is dilated by a scale factor of 2.5.

How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q2/5

1205

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q2/5

A dilation is applied to a line that passes through the center of dilation.

What is the image?

A. A parallel line

B. A single point

C. A perpendicular line

D. The same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q3/5

1206

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. The same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q3/5

True or false?

A dilation maps a line that does not pass through the center to a parallel line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q4/5

1207

QUESTION ID

S39-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q4/5

True or false?

A dilation with a scale factor of 2 doubles every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q5/5

1208

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Dilations keep angle measures the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.1 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 8.G.A.3

FORWARD 2

Use similar triangles to solve problems • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 6-foot person casts a 4-foot shadow. At the same time, a tree casts a 22-foot shadow.

How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q1/5

1210

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

33 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q1/5

Triangle ABC is similar to triangle DEF, with AB = 8, DE = 12, and BC = 10.

Find EF.

EF = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q2/5

1211

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q2/5

A segment parallel to one side of a triangle splits a second side into parts of 3 and 6. It splits the third side into a first part of 4 and a second part.

How long is the second part?

A. 2

B. 12

C. 8

D. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q3/5

1212

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S39-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q3/5

True or false?

In similar triangles, the ratio of the perimeters equals the scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q4/5

1213

QUESTION ID

S39-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q4/5

True or false?

Triangles with sides 3, 4, 5 and 6, 8, 12 are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q5/5

1214

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S39-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 39 • FORWARD 2 • Q5/5

SET 40 • 8.G.A.4 • GEOMETRY

Describe a sequence of transformations that shows two figures are similar

MAIN	Grade 8 main question	**8.G.A.4**
BACKWARD 1	Find a scale factor between scale copies	**7.G.A.1**
BACKWARD 2	Decide whether two ratios are equivalent	**7.RP.A.2.a**
FORWARD 1	Use similarity transformations to decide whether triangles are sim…	**HSG-SRT.A.2**
FORWARD 2	Use a translation and a dilation to show that any two circles are s…	**HSG-C.A.1**

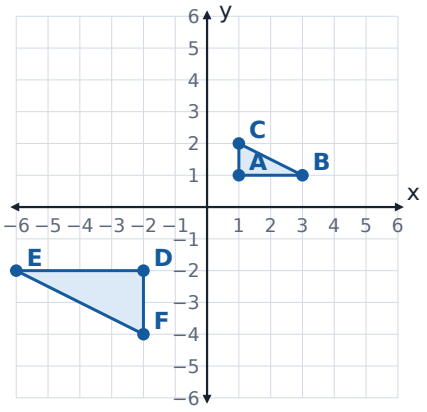
SET 40 • 8.G.A.4

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.4

ANSWER KEY Reduced copy of the question page, with its answer

Describe a sequence of transformations that maps triangle ABC onto triangle DEF.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q1/5

1217

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

A dilation with a scale factor of 2 and the origin as the center, followed by a 180° rotation about the origin.

GRADING NOTE Any correct sequence is accepted, for example the rotation first, then the dilation.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q1

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q1/5

Explain why a 2-by-3 rectangle and a 4-by-6 rectangle are similar.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q2/5

1218

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

A dilation with a scale factor of 2 (followed by rigid motions to move it into place) maps the first rectangle onto the second.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q2/5

Which sequence of transformations maps a figure onto a figure that is similar but not congruent?

A. A rotation, then a translation

B. A translation only

C. Two reflections

D. A dilation with a scale factor of 3, then a translation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q3/5

1219

QUESTION ID

S40-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A dilation with a scale factor of 3, then a translation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q3/5

Are a 3-4-5 triangle and a 5-12-13 triangle similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q4/5

1220

QUESTION ID

S40-M-Q4

TYPE

Short answer

CORRECT ANSWER

No. The ratios of corresponding sides (3/5, 4/12, 5/13) are not equal, so no dilation and rigid motions can map one onto the other.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-M-Q4

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q4/5

True or false?

Two figures are similar if a sequence of rotations, reflections, translations, and dilations maps one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q5/5

1221

QUESTION ID

S40-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 8 • Geometry • 8.G.A.4 • Set 40 • MAIN • Q5/5

SET 40 • 8.G.A.4

BACKWARD 1

Find a scale factor between scale copies • Grade 7 • Geometry • 7.G.A.1

A 3 cm side becomes 12 cm in a scale copy.

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q1/5

1223

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q1/5

A 15 m side becomes 5 m in a scale copy.

What is the scale factor?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q2/5

1224

QUESTION ID

S40-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q2/5

A triangle's sides are 2, 3, and 4. Which side lengths are a scale copy?

A. 6, 9, 12

B. 4, 5, 6

C. 2, 6, 8

D. 3, 4, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q3/5

1225

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6, 9, 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q3/5

True or false?

A scale copy with a scale factor of 1 is the same size as the original.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q4/5

1226

QUESTION ID

S40-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q4/5

True or false?

A rectangle 2 by 3 and a rectangle 3 by 4 are scale copies of each other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q5/5

1227

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 8.G.A.4

BACKWARD 2

Decide whether two ratios are equivalent • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Are 4 : 6 and 10 : 15 equivalent ratios?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q1/5

1229

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; both equal 2 : 3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q1/5

Are 3 : 5 and 9 : 12 equivalent ratios?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q2/5

1230

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

No; 3/5 = 0.6 and 9/12 = 0.75.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q2/5

Which ratio is equivalent to 5 : 8?

A. 25 : 48

B. 15 : 24

C. 8 : 5

D. 10 : 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q3/5

1231

QUESTION ID

S40-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 15 : 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-B2-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q3/5

True or false?

2 : 7 and 6 : 21 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q4/5

1232

QUESTION ID

S40-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q4/5

True or false?

4 : 5 and 5 : 6 are equivalent ratios.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q5/5

1233

QUESTION ID

S40-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 8.G.A.4

FORWARD 1

Use similarity transformations to decide whether triangles are similar • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has sides 5, 7, and 9. Triangle DEF has sides 10, 14, and 18.

Are they similar? If so, what is the scale factor from ABC to DEF?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q1/5

1235

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; the scale factor is 2.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q1/5

Two triangles are similar. Explain what is true about their corresponding angles and sides.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q2/5

1236

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Corresponding angles are equal, and corresponding sides are proportional.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q2/5

Triangle PQR is similar to triangle STU with a scale factor of 3. PQ = 4.

What is ST?

A. 12

B. 4

C. $\frac{4}{3}$

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q3/5

1237

QUESTION ID

S40-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q3/5

True or false?

If a dilation followed by a rotation maps one triangle onto another, the triangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q4/5

1238

QUESTION ID

S40-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q4/5

True or false?

Any two rectangles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q5/5

1239

QUESTION ID

S40-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 40 • FORWARD 1 • Q5/5

SET 40 • 8.G.A.4

FORWARD 2

Use a translation and a dilation to show that any two circles are similar • High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1

Circle A has center (0, 0) and radius 2. Circle B has center (5, 1) and radius 6.

Describe a translation followed by a dilation that maps circle A onto circle B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q1/5

1241

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Translate 5 units right and 1 unit up, then dilate about (5, 1) with a scale factor of 3.

GRADING NOTE Any correct sequence is accepted, for example a dilation about (0, 0) with scale factor 3 followed by a translation 5 units right and 1 unit up.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q1

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q1/5

What scale factor of a dilation about the center maps a circle with a radius of 4 onto a circle with the same center and a radius of 10?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q2/5

1242

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2.5

GRADING NOTE $\frac{5}{2}$ is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q2

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q2/5

Which statement explains why any two circles are similar?

A. A rotation alone maps any circle onto any other circle.

B. All circles have the same radius.

C. A translation moves one center onto the other, and then a dilation about that center makes the radii equal.

D. All circles have the same area.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q3/5

1243

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. A translation moves one center onto the other, and then a dilation about that center makes the radii equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q3

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q3/5

True or false?

Any two circles are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q4/5

1244

QUESTION ID

S40-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q4/5

True or false?

A circle with a radius of 3 can be mapped onto a circle with a radius of 6 by a translation alone.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q5/5

1245

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE A translation keeps the radius 3; a dilation is also needed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S40-F2-Q5

High School (Grade 10 in this bank) • Geometry: Circles • HSG-C.A.1 • Set 40 • FORWARD 2 • Q5/5

SET 41 • 8.G.A.5 • GEOMETRY

Use the angle sum and exterior angles of a triangle

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Write and solve equations for supplementary and complementary…	**7.G.B.5**
BACKWARD 2	Find an unknown angle in a sum of angles	**4.MD.C.7**
FORWARD 1	Prove theorems about triangles	**HSG-CO.C.10**
FORWARD 2	Define trigonometric ratios from similar right triangles	**HSG-SRT.C.6**

SET 41 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

ANSWER KEY Reduced copy of the question page, with its answer

Two angles of a triangle measure 52° and 71°.

What does the third angle measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q1/5

1248

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

57°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q1/5

The angles of a triangle measure x°, (x + 10)°, and (2x − 30)°.

Write and solve an equation to find x.

Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q2/5

1249

QUESTION ID

S41-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation x + (x + 10) + (2x − 30) = 180 and the answer x = 50

GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q2/5

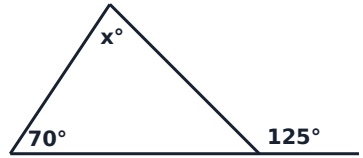
In the figure, what is the value of x?

A. 55

B. 70

C. 125

D. 35



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q3/5

1250

QUESTION ID

S41-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q3/5

Explain why an exterior angle of a triangle equals the sum of the two remote interior angles. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q4/5

1251

QUESTION ID

S41-M-Q4

TYPE

Answer with required work

CORRECT ANSWER

The exterior angle = the sum of the two remote interior angles

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: the exterior angle and the adjacent interior angle form a linear pair, so they add to 180°; the three interior angles also add to 180°; so the exterior angle equals the sum of the other two interior angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q5/5

1252

QUESTION ID

S41-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Two obtuse angles alone add to more than 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 41 • MAIN • Q5/5

SET 41 • 8.G.A.5

BACKWARD 1

Write and solve equations for supplementary and complementary angles • Grade 7 • Geometry • 7.G.B.5

Two supplementary angles measure x° and 130°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q1/5

1254

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q1/5

Two complementary angles measure 2x° and 40°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q2/5

1255

QUESTION ID

S41-B1-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q2/5

Angles of x° and (x + 30)° are supplementary.

What is x?

A. 75

B. 150

C. 105

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q3/5

1256

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q3/5

True or false?

Angles of 47° and 43° are complementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q4/5

1257

QUESTION ID

S41-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q4/5

True or false?

Angles of 100° and 90° are supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q5/5

1258

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 8.G.A.5

BACKWARD 2

Find an unknown angle in a sum of angles • Grade 4 • Measurement & Data • 4.MD.C.7

A 90° angle is split into two angles. One is 35°.

What is the other?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q1/5

1260

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

55°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q1/5

A straight angle (180°) is split into two angles. One is 110°.

What is the other?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q2/5

1261

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

70°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q2/5

An angle of 120° is split into angles of 75° and x°.

What is x?

A. 195

B. 55

C. 45

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q3/5

1262

QUESTION ID

S41-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q3/5

True or false?

Angles of 30° and 60° together make a 90° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q4/5

1263

QUESTION ID

S41-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q4/5

True or false?

Angles of 80° and 80° together make a 180° angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q5/5

1264

QUESTION ID

S41-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 41 • BACKWARD 2 • Q5/5

SET 41 • 8.G.A.5

FORWARD 1

Prove theorems about triangles • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

ANSWER KEY Reduced copy of the question page, with its answer

Prove that the angles of triangle ABC add to 180°. Use a line through C that is parallel to AB. Give a reason for each step.

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q1/5

1266

QUESTION ID

S41-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle A + angle B + angle C = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: draw line l through C parallel to AB. The angles at C along l add to 180° (straight angle). The two outer ones equal angles A and B (alternate interior angles, l ∥ AB). So angle A + angle B + angle C = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q1/5

The angles of a triangle are 2x°, 3x°, and 4x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q2/5

1267

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q2/5

An isosceles triangle has base angles of 40°.

What does its third angle measure?

A. 100°

B. 40°

C. 80°

D. 140°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q3/5

1268

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 100°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q3/5

True or false?

The base angles of an isosceles triangle are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q4/5

1269

QUESTION ID

S41-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q4/5

True or false?

An exterior angle of a triangle equals the sum of all three interior angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q5/5

1270

QUESTION ID

S41-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 41 • FORWARD 1 • Q5/5

SET 41 • 8.G.A.5

FORWARD 2

Define trigonometric ratios from similar right triangles • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is sin A?

sin A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q1/5

1272

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q1/5

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is tan A?

tan A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q2/5

1273

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q2/5

A right triangle has a leg of 3 opposite angle A, a leg of 4 next to angle A, and a hypotenuse of 5.

What is cos A?

A. $\frac{3}{5}$

B. $\frac{5}{4}$

C. $\frac{4}{5}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q3/5

1274

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S41-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q3/5

True or false?

In every right triangle with a 30° angle, the side opposite the 30° angle is half the hypotenuse, so $\sin 30° = \frac{1}{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q4/5

1275

QUESTION ID

S41-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q4/5

True or false?

sin A changes when a right triangle is enlarged but its angles stay the same.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q5/5

1276

QUESTION ID

S41-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Similar right triangles have equal ratios.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S41-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.6 • Set 41 • FORWARD 2 • Q5/5

SET 42 • 8.G.A.5 • GEOMETRY

Write and solve equations for angles formed by parallel lines cut by a transversal

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Write and solve equations for vertical and adjacent angles	**7.G.B.5**
BACKWARD 2	Solve two-step equations	**7.EE.B.4.a**
FORWARD 1	Prove that alternate interior angles are congruent	**HSG-CO.C.9**
FORWARD 2	Use slopes to show that lines are parallel or perpendicular	**HSG-GPE.B.5**

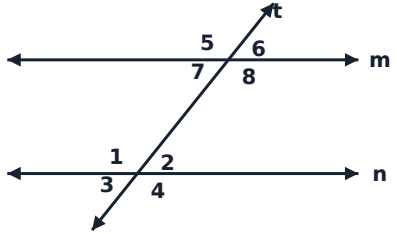
SET 42 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

In the figure, lines m and n are parallel and angle 2 = 52°.

What does angle 6 measure? Give the reason.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q1/5

1279

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

52°; corresponding angles are congruent.

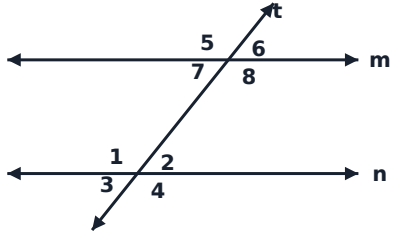
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Angle 1 = (3x + 10)° and angle 8 = 115°.

Write and solve an equation for x.



Equation and answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q2/5

1280

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

The equation 3x + 10 = 115 (alternate interior angles are congruent) and the answer x = 35

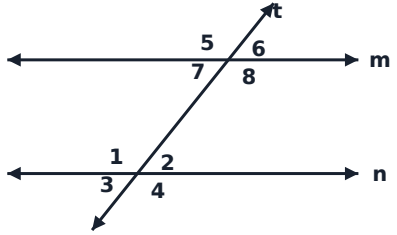
GRADING NOTE Both parts are required: a valid equation and the correct solution. Any valid equation is accepted, but the solution must also be correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q2/5

In the figure, lines m and n are parallel. Angle 2 = (2x + 4)° and angle 8 = (4x − 10)°.

What is x?



A. 93 **B.** 31

C. 43 **D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q3/5

1281

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

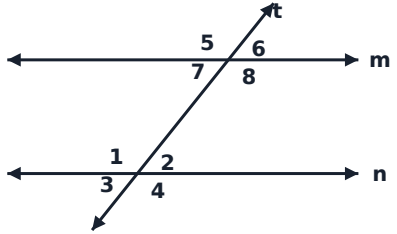
B. 31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q3/5

In the figure, lines m and n are parallel and angle 4 = 128°.

What does angle 5 measure? Give the reasons.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q4/5

1282

QUESTION ID

S42-M-Q4

TYPE

Short answer

CORRECT ANSWER

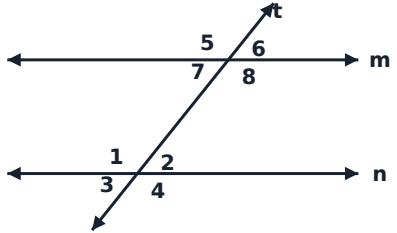
128°: angle 1 = angle 4 (vertical angles), and angle 5 = angle 1 (corresponding angles).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q4/5

True or false?

In the figure, angles 3 and 6 are alternate exterior angles, so they are congruent when m and n are parallel.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q5/5

1283

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 42 • MAIN • Q5/5

SET 42 • 8.G.A.5

BACKWARD 1

Write and solve equations for vertical and adjacent angles • Grade 7 • Geometry • 7.G.B.5

Two lines intersect. Vertical angles measure (2x + 20)° and 80°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q1/5

1285

QUESTION ID

S42-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q1/5

Two lines intersect. Adjacent angles measure x° and 3x°.

What is x?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q2/5

1286

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q2/5

Two lines intersect. Adjacent angles measure (x + 40)° and 2x°.

What is x?

A. $\frac{140}{3}$

B. 50

C. 40

D. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q3/5

1287

QUESTION ID

S42-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 140/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q3/5

True or false?

Vertical angles of 5x° and 75° give x = 15.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q4/5

1288

QUESTION ID

S42-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q4/5

True or false?

Adjacent angles that form a straight line measure (x + 10)° and x°. Then x = 90.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q5/5

1289

QUESTION ID

S42-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE x = 85.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 8.G.A.5

BACKWARD 2

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

3x + 10 = 115

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q1/5

1291

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q1/5

Solve for x.

$2x - 15 = 75$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q2/5

1292

QUESTION ID

S42-B2-Q2

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Solve for x.

$6x - 6 = 180$

A. x = 174

B. x = 29

C. x = 31

D. x = 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q3/5

1293

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. x = 31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q3/5

True or false?

x = 20 is the solution of 4x + 20 = 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q4/5

1294

QUESTION ID

S42-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$x = 25$ is the solution of $5x - 5 = 115$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q5/5

1295

QUESTION ID

S42-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is $x = 24$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-B2-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • BACKWARD 2 • Q5/5

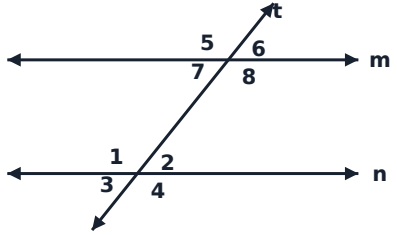
SET 42 • 8.G.A.5

FORWARD 1

Prove that alternate interior angles are congruent • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 1 = angle 8. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q1/5

1297

QUESTION ID

S42-F1-Q1

TYPE

Answer with required work

CORRECT ANSWER

Angle 1 = angle 8

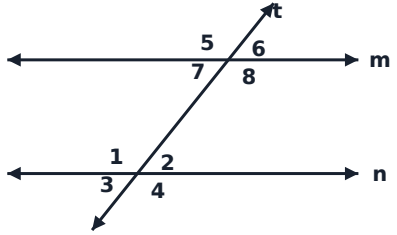
GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 1 = angle 5 (corresponding angles, m ∥ n); angle 5 = angle 8 (vertical angles); so angle 1 = angle 8.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

In the figure, lines m and n are parallel. Prove that angle 2 + angle 8 = 180°. Give a reason for each step.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q2/5

1298

QUESTION ID

S42-F1-Q2

TYPE

Answer with required work

CORRECT ANSWER

Angle 2 + angle 8 = 180°

GRADING NOTE REQUIRED WORK: A valid chain of reasons. Other valid orders are accepted. A conclusion without reasons is not full credit. Example: angle 6 + angle 8 = 180° (linear pair); angle 2 = angle 6 (corresponding angles); so angle 2 + angle 8 = 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q2/5

A proof says: angle 1 = angle 5 (corresponding angles). Angle 5 = angle 8 (?). So angle 1 = angle 8.

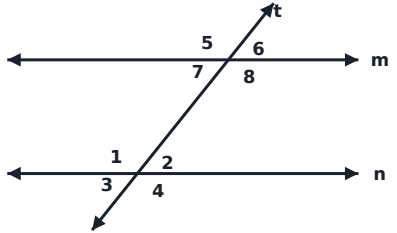
What is the missing reason?

A. Linear pairs are supplementary.

B. Vertical angles are congruent.

C. Alternate interior angles are congruent.

D. The triangle angle sum is 180°.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q3/5

1299

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Vertical angles are congruent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q3/5

True or false?

If corresponding angles formed by a transversal are congruent, then the two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q4/5

1300

QUESTION ID

S42-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q4/5

1300

True or false?

"The angles look equal in the figure" is a valid reason in a proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q5/5

1301

QUESTION ID

S42-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.9 • Set 42 • FORWARD 1 • Q5/5

SET 42 • 8.G.A.5

FORWARD 2

Use slopes to show that lines are parallel or perpendicular • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

A transversal crosses the lines $y = 2x + 1$ and $y = 2x - 3$.

Are the two lines parallel? Explain with slopes.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q1/5

1303

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes; both lines have slope 2 and different y-intercepts.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q1/5

Write an equation of the line through (1, 4) that is perpendicular to $y = -\frac{1}{2}x + 3$.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q2/5

1304

QUESTION ID

S42-F2-Q2

TYPE

Short answer

CORRECT ANSWER

y = 2x + 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q2/5

Which pair of lines is perpendicular?

A. $y = x$ and $y = x + 2$

B. $y = 3x$ and $y = -3x$

C. $y = 3x + 1$ and $y = 3x - 1$

D. $y = 3x + 1$ and $y = -\frac{1}{3}x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q3/5

1305

QUESTION ID

S42-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $y = 3x + 1$ and $y = -(1/3)x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S42-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q3/5

True or false?

Two non-vertical lines are parallel exactly when their slopes are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q4/5

1306

QUESTION ID

S42-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q4/5

True or false?

The lines $y = \frac{2}{5}x$ and $y = \frac{5}{2}x$ are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q5/5

1307

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Perpendicular slopes multiply to −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S42-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 42 • FORWARD 2 • Q5/5

SET 43 • 8.G.A.5 • GEOMETRY

Use the angle-angle criterion to show that triangles are similar

MAIN	Grade 8 main question	**8.G.A.5**
BACKWARD 1	Find a missing length in a scale copy	**7.G.A.1**
BACKWARD 2	Use the triangle angle sum to find a missing angle	**7.G.B.5**
FORWARD 1	Establish the angle-angle criterion with similarity transformations	**HSG-SRT.A.3**
FORWARD 2	Use similar triangles to solve problems	**HSG-SRT.B.5**

SET 43 • 8.G.A.5

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.A.5

Triangle ABC has angles of 35° and 85°. Triangle DEF has angles of 35° and 60°.

Are the triangles similar? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q1/5

1310

QUESTION ID

S43-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The third angle of DEF is 180° − 35° − 60° = 85°, so two pairs of angles are equal (AA).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q1/5

A 5-foot post casts a 3-foot shadow. At the same time, a tree casts a 12-foot shadow.

Explain why the two triangles are similar, and find the height of the tree.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q2/5

1311

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

Both triangles have a right angle and the same angle of the sun's rays, so they are similar by AA. The tree is 20 feet tall.

GRADING NOTE Grade both: the AA reason and 20 feet.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-M-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q2/5

Which information shows that two triangles are similar by the angle-angle criterion?

A. One pair of sides is proportional.

B. All the sides are equal.

C. Two pairs of corresponding angles are equal.

D. One pair of corresponding angles is equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q3/5

1312

QUESTION ID

S43-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Two pairs of corresponding angles are equal.

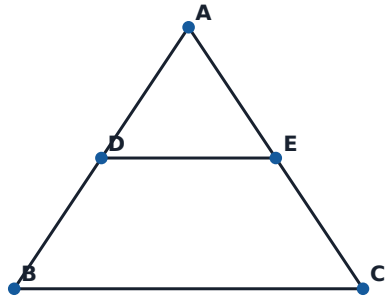
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q3/5

In the figure, segment DE is parallel to BC.

Explain why triangle ADE is similar to triangle ABC.



Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q4/5

1313

QUESTION ID

S43-M-Q4

TYPE

Short answer

CORRECT ANSWER

The triangles share angle A, and angle ADE = angle ABC (corresponding angles, because DE ∥ BC). So they are similar by AA.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q4/5

True or false?

Two triangles that each have a 90° angle and a 40° angle are similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q5/5

1314

QUESTION ID

S43-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 43 • MAIN • Q5/5

SET 43 • 8.G.A.5

BACKWARD 1

Find a missing length in a scale copy • Grade 7 • Geometry • 7.G.A.1

A triangle has sides 4, 6, and 8. In a scale copy, the side of 4 becomes 10.

What does the side of 6 become?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

1316

QUESTION ID

S43-B1-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q1/5

A drawing with a scale factor of 3 shows a 2 cm side.

How long is that side in the scale copy?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

1317

QUESTION ID

S43-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q2/5

A 5-by-8 rectangle is scaled so the 5 becomes 15.

What does the 8 become?

A. 40

B. 24

C. 13

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

1318

QUESTION ID

S43-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q3/5

True or false?

In a scale copy, every length is multiplied by the same scale factor.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

1319

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q4/5

True or false?

In a scale copy with a scale factor of 2, every angle is doubled.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

1320

QUESTION ID

S43-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 8.G.A.5

BACKWARD 2

Use the triangle angle sum to find a missing angle • Grade 7 • Geometry • 7.G.B.5

A triangle has angles of 35° and 60°.

What is the third angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q1/5

1322

QUESTION ID

S43-B2-Q1

TYPE

Short answer

CORRECT ANSWER

85°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q1/5

A right triangle has a 40° angle.

What is the third angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q2/5

1323

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

50°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q2/5

A triangle has angles of 90° and 25°.

What is the third angle?

A. 75°

B. 65°

C. 115°

D. 55°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q3/5

1324

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 65°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q3/5

True or false?

A triangle with angles of 50° and 60° has a third angle of 70°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q4/5

1325

QUESTION ID

S43-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q4/5

True or false?

A triangle with angles of 45° and 45° has a third angle of 45°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q5/5

1326

QUESTION ID

S43-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 8.G.A.5

FORWARD 1

Establish the angle-angle criterion with similarity transformations • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3

Triangles ABC and DEF have angle A = angle D and angle B = angle E.

Which transformations show that the triangles are similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q1/5

1328

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

A dilation that makes AB the same length as DE, followed by rigid motions that move the image onto DEF; the equal angles make the third vertices match.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q1/5

Why is it enough to know two pairs of equal angles to show triangles are similar?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q2/5

1329

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

The third angles must also be equal, because each triangle's angles add to 180°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q2/5

Triangles with angles 30°, 70°, 80° and 70°, 30°, 80° are...

A. congruent in every case

B. similar

C. not similar

D. impossible to compare

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q3/5

1330

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. similar

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q3/5

True or false?

Two equilateral triangles are always similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q4/5

1331

QUESTION ID

S43-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q4/5

True or false?

Two isosceles triangles are always similar.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q5/5

1332

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.3 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 8.G.A.5

FORWARD 2

Use similar triangles to solve problems • High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

A 1.5 m stick casts a 2 m shadow. A building casts a 36 m shadow at the same time.

How tall is the building?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q1

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

1334

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

27 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q1/5

Triangle PQR is similar to triangle XYZ with PQ = 6, XY = 9, and QR = 8.

Find YZ.

YZ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

1335

QUESTION ID

S43-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q2

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q2/5

Two similar triangles have a scale factor of 3. The smaller one has a perimeter of 14.

What is the larger one's perimeter?

A. 17

B. 14

C. 42

D. 126

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q3

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

1336

QUESTION ID

S43-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S43-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q3/5

True or false?

A mirror can be used with similar triangles to find the height of a tall object.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

1337

QUESTION ID

S43-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q4

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q4/5

True or false?

If two triangles are similar with a scale factor of 2, their areas are also in the ratio 2 : 1.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

1338

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Their areas are in the ratio 4 : 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S43-F2-Q5

High School (Grade 10 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 43 • FORWARD 2 • Q5/5

SET 44 • 8.G.B.6 • GEOMETRY

Explain a proof of the Pythagorean theorem and its converse

MAIN	Grade 8 main question	**8.G.B.6**
BACKWARD 1	Find the area of a right triangle	**6.G.A.1**
BACKWARD 2	Find the area of a square	**3.MD.C.7.b**
FORWARD 1	Prove the Pythagorean theorem with similar triangles	**HSG-SRT.B.4**
FORWARD 2	Derive the equation of a circle from the Pythagorean theorem	**HSG-GPE.A.1**

SET 44 • 8.G.B.6

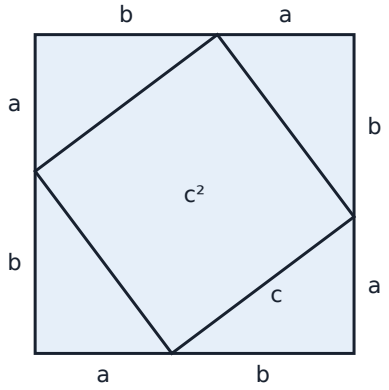
MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.6

ANSWER KEY Reduced copy of the question page, with its answer

The figure shows a square with sides a + b, made of four right triangles and a tilted square with side c.

Explain how the figure proves that $a^2 + b^2 = c^2$.



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q1/5

1341

QUESTION ID

S44-M-Q1

TYPE

Answer with required work

CORRECT ANSWER

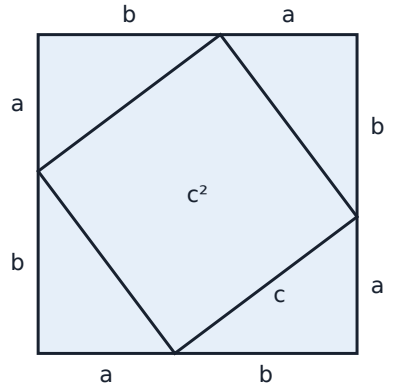
$a^2 + b^2 = c^2$

GRADING NOTE REQUIRED WORK: A complete argument: the large square's area is $(a + b)^2 = a^2 + 2ab + b^2$; it is also 4 triangles ($4 \cdot \frac{1}{2}ab = 2ab$) plus the inner square c^2; so $a^2 + 2ab + b^2 = 2ab + c^2$, and subtracting 2ab gives $a^2 + b^2 = c^2$. Any equivalent wording is accepted.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q1/5

In the figure, write the area of the large square as the sum of the areas of the four triangles and the inner square.



Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q2/5

1342

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

4 ·

$$\frac{1}{2}ab + c^2 = 2ab + c^2$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q2/5

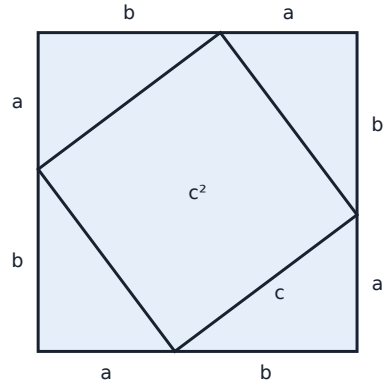
In the proof with the figure, which step comes after writing $(a + b)^2 = 2ab + c^2$?

A. Divide both sides by c

B. Expand to $a^2 + 2ab + b^2 = 2ab + c^2$, then subtract 2ab from both sides

C. Add 2ab to both sides

D. Take the square root of each term



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q3/5

1343

QUESTION ID

S44-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Expand to $a^2 + 2ab + b^2 = 2ab + c^2$, then subtract 2ab from both sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q3/5

A triangle has sides a, b, and c with $a^2 + b^2 = c^2$. Explain why it must be a right triangle.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q4/5

1344

QUESTION ID

S44-M-Q4

TYPE

Short answer

CORRECT ANSWER

A right triangle with legs a and b has a hypotenuse of $\sqrt{(a^2 + b^2)} = c$. So the two triangles have the same three sides and are congruent (SSS), and the given triangle has a right angle too.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q4/5

1344

True or false?

The Pythagorean theorem $a^2 + b^2 = c^2$ is true for every triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q5/5

1345

QUESTION ID

S44-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is true only for right triangles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 44 • MAIN • Q5/5

SET 44 • 8.G.B.6

BACKWARD 1

Find the area of a right triangle • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs of 3 and 4.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q1/5

1347

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q1/5

A right triangle has legs of 5 and 12.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q2/5

1348

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

30 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q2/5

What is the total area of four right triangles that each have legs of 2 and 6?

A. 24

B. 12

C. 48

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q3/5

1349

QUESTION ID

S44-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q3/5

True or false?

A right triangle with legs of 8 and 3 has an area of 12 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q4/5

1350

QUESTION ID

S44-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q4/5

True or false?

A right triangle with legs of 6 and 6 has an area of 36 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q5/5

1351

QUESTION ID

S44-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 8.G.B.6

BACKWARD 2

Find the area of a square • Grade 3 • Measurement & Data • 3.MD.C.7.b

A square has sides of 7 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q1/5

1353

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

49 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q1/5

A square has sides of 9 units.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q2/5

1354

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

81 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q2/5

A square has sides of 5 units.

What is its area?

A. 15 square units

B. 25 square units

C. 10 square units

D. 20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q3/5

1355

QUESTION ID

S44-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 25 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q3/5

True or false?

A square with 4-unit sides has an area of 16 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q4/5

1356

QUESTION ID

S44-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q4/5

True or false?

A square with 6-unit sides has an area of 24 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q5/5

1357

QUESTION ID

S44-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 44 • BACKWARD 2 • Q5/5

SET 44 • 8.G.B.6

FORWARD 1

Prove the Pythagorean theorem with similar triangles • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4

In right triangle ABC with the right angle at C, altitude CD is drawn to hypotenuse AB.

Why is $AC^2 = AD \cdot AB$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q1/5

1359

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Triangle ACD is similar to triangle ABC (AA), so $AC/AB = AD/AC$, which gives $AC^2 = AD \cdot AB$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q1/5

In the same triangle, $BC^2 = BD \cdot AB$. Add this to $AC^2 = AD \cdot AB$. What do you get?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q2/5

1360

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$AC^2 + BC^2 = AD \cdot AB + BD \cdot AB = AB(AD + BD) = AB \cdot AB = AB^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q2/5

Which criterion shows that triangle ACD is similar to triangle ABC?

A. ASA

B. AA

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q3/5

1361

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. AA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q3/5

True or false?

The altitude to the hypotenuse divides a right triangle into two triangles similar to it.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q4/5

1362

QUESTION ID

S44-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q4/5

True or false?

The altitude to the hypotenuse always cuts the hypotenuse into two equal parts.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q5/5

1363

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.4 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 8.G.B.6

FORWARD 2

Derive the equation of a circle from the Pythagorean theorem • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1

Write the equation of the circle with center (0, 0) and radius 5.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q1/5

1365

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x^2 + y^2 = 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q1/5

Write the equation of the circle with center (2, −1) and radius 3.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q2/5

1366

QUESTION ID

S44-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$(x - 2)^2 + (y + 1)^2 = 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q2/5

Which point is on the circle $x^2 + y^2 = 100$?

A. (5, 5)

B. (10, 10)

C. (6, 8)

D. (6, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q3/5

1367

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (6, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S44-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q3/5

True or false?

The equation $(x - h)^2 + (y - k)^2 = r^2$ comes from the Pythagorean theorem.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q4/5

1368

QUESTION ID

S44-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q4/5

True or false?

The circle $x^2 + y^2 = 16$ has a radius of 16.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q5/5

1369

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its radius is 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S44-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.A.1 • Set 44 • FORWARD 2 • Q5/5

SET 45 • 8.G.B.6 • GEOMETRY

Use the converse of the Pythagorean theorem to decide whether a triangle is a right triangle

MAIN	Grade 8 main question	**8.G.B.6**
BACKWARD 1	Find the squares of whole numbers	**6.EE.A.1**
BACKWARD 2	Evaluate an expression with exponents	**6.EE.A.2.c**
FORWARD 1	Classify a triangle as acute, right, or obtuse from its side lengths	**HSG-CO.C.10**
FORWARD 2	Use coordinates to show that a triangle is a right triangle	**HSG-GPE.B.4**

SET 45 • 8.G.B.6

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.6

Is a triangle with sides 9, 12, and 15 a right triangle? Show your check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q1/5

1372

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

Yes.

$9^2 + 12^2 = 81 + 144 = 225 = 15^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q1/5

Is a triangle with sides 7, 8, and 11 a right triangle? Show your check.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q2/5

1373

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

No. $7^2 + 8^2 = 49 + 64 = 113$, but $11^2 = 121$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q2/5

Which side lengths form a right triangle?

A. 6, 8, 12

B. 20, 21, 29

C. 5, 7, 9

D. 10, 12, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q3/5

1374

QUESTION ID

S45-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 20, 21, 29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q3/5

A carpenter checks a corner. Points 3 feet and 4 feet from the corner along the two edges are 5 feet apart.

Is the corner a right angle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q4/5

1375

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

Yes. $3^2 + 4^2 = 5^2$, so the angle is 90°.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q4/5

True or false?

A triangle with sides 2, 3, and 4 is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q5/5

1376

QUESTION ID

S45-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4 + 9 = 13$, but $4^2 = 16$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-M-Q5

Grade 8 • Geometry • 8.G.B.6 • Set 45 • MAIN • Q5/5

SET 45 • 8.G.B.6

BACKWARD 1

Find the squares of whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

15^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q1/5

1378

QUESTION ID

S45-B1-Q1

TYPE

Short answer

CORRECT ANSWER

225

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q1/5

Find the value.

11^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q2/5

1379

QUESTION ID

S45-B1-Q2

TYPE

Short answer

CORRECT ANSWER

121

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

13^2

A. 26

B. 169

C. 139

D. 196

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q3/5

1380

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 169

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q3/5

True or false?

$20^2 = 400$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q4/5

1381

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q4/5

True or false?

$9^2 = 18$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q5/5

1382

QUESTION ID

S45-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 8.G.B.6

BACKWARD 2

Evaluate an expression with exponents • Grade 6 • Expressions & Equations • 6.EE.A.2.c

ANSWER KEY Reduced copy of the question page, with its answer

Evaluate $a^2 + b^2$ when a = 6 and b = 8.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q1/5

1384

QUESTION ID

S45-B2-Q1

TYPE

Short answer

CORRECT ANSWER

100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q1/5

Evaluate $a^2 + b^2$ when a = 5 and b = 12.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q2/5

1385

QUESTION ID

S45-B2-Q2

TYPE

Short answer

CORRECT ANSWER

169

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q2/5

Evaluate $c^2 - a^2$ when $c = 10$ and $a = 6$.

A. 16

B. 136

C. 64

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q3/5

1386

QUESTION ID

S45-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q3/5

True or false?

When a = 3 and b = 4, $a^2 + b^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q4/5

1387

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q4/5

True or false?

When a = 2 and b = 3, $a^2 + b^2 = 25$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q5/5

1388

QUESTION ID

S45-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 8.G.B.6

FORWARD 1

Classify a triangle as acute, right, or obtuse from its side lengths • High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10

Is a triangle with sides 5, 6, and 8 acute, right, or obtuse? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q1/5

1390

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Obtuse; $5^2 + 6^2 = 61$, which is less than $8^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q1

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q1/5

Is a triangle with sides 6, 7, and 8 acute, right, or obtuse? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q2/5

1391

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Acute; $6^2 + 7^2 = 85$, which is greater than $8^2 = 64$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q2

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q2/5

A triangle has sides 8, 15, and 17.

What kind of triangle is it?

A. Obtuse

B. Acute

C. Right

D. Not a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q3/5

1392

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q3

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q3/5

True or false?

If $a^2 + b^2 < c^2$ for the longest side c, the triangle is obtuse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q4/5

1393

QUESTION ID

S45-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F1-Q4

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q4/5

True or false?

A triangle with sides 4, 4, and 7 is acute.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q5/5

1394

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE $4^2 + 4^2 = 32 < 49$, so it is obtuse.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION — ID S45-F1-Q5

High School (Grade 9 in this bank) • Geometry: Congruence • HSG-CO.C.10 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 8.G.B.6

FORWARD 2

Use coordinates to show that a triangle is a right triangle • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4

Is the triangle with vertices (1, 1), (4, 1), and (4, 5) a right triangle? Use side lengths.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q1/5

1396

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Yes. The sides are 3, 4, and 5, and $3^2 + 4^2 = 5^2$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q1/5

Use slopes to decide whether the triangle with vertices (0, 0), (2, 4), and (6, 2) has a right angle at (2, 4).

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q2/5

1397

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. The slope from (2, 4) to (0, 0) is 2 and the slope from (2, 4) to (6, 2) is −1/2; their product is −1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q2/5

Which triangle is a right triangle?

A. (0, 0), (2, 2), (4, 1)

B. (0, 0), (3, 1), (1, 3)

C. (1, 1), (2, 3), (3, 1)

D. (0, 0), (3, 0), (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q3/5

1398

QUESTION ID

S45-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (0, 0), (3, 0), (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S45-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q3/5

True or false?

The triangle with vertices (0, 0), (5, 0), and (0, 5) is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q4/5

1399

QUESTION ID

S45-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q4/5

True or false?

The triangle with vertices (0, 0), (4, 0), and (2, 1) is a right triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q5/5

QUESTION ID

S45-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S45-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.4 • Set 45 • FORWARD 2 • Q5/5

SET 46 • 8.G.B.7 • GEOMETRY

Use the Pythagorean theorem to find unknown lengths in two dimensions

MAIN	Grade 8 main question	**8.G.B.7**
BACKWARD 1	Find the squares of decimals and whole numbers	**6.EE.A.1**
BACKWARD 2	Solve one-step equations	**6.EE.B.7**
FORWARD 1	Solve right triangles with the Pythagorean theorem and trigonom...	**HSG-SRT.C.8**
FORWARD 2	Apply geometric methods to solve design problems	**HSG-MG.A.3**

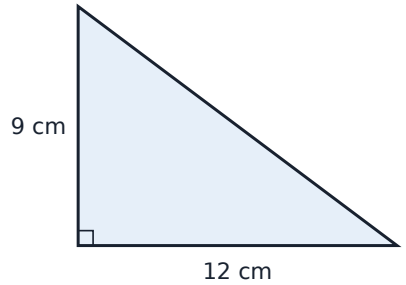
SET 46 • 8.G.B.7

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.7

Find the length of the hypotenuse.

Hypotenuse: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q1/5

1403

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

15 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q1/5

A 13-foot ladder leans against a wall. Its base is 5 feet from the wall.

How high up the wall does it reach?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q2/5

1404

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

12 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q2/5

A rectangular screen is 48 inches wide and 36 inches tall.

How long is its diagonal?

A. 60 inches

B. 84 inches

C. 72 inches

D. 42 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q3/5

1405

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 60 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q3/5

A right triangle has a hypotenuse of 17 and a leg of 8.

How long is the other leg?

Leg: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q4/5

1406

QUESTION ID

S46-M-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q4/5

True or false?

A square with 10 cm sides has a diagonal of about 14.1 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q5/5

1407

QUESTION ID

S46-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 46 • MAIN • Q5/5

SET 46 • 8.G.B.7

BACKWARD 1

Find the squares of decimals and whole numbers • Grade 6 • Expressions & Equations • 6.EE.A.1

Find the value.

14^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q1/5

1409

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

196

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q1/5

Find the value.

2.5^2

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q2/5

1410

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Find the value.

17^2

A. 189

B. 279

C. 289

D. 34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q3/5

1411

QUESTION ID

S46-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 289

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q3/5

True or false?

$9^2 = 81$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q4/5

1412

QUESTION ID

S46-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

$1.5^2 = 3$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q5/5

1413

QUESTION ID

S46-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 8.G.B.7

BACKWARD 2

Solve one-step equations • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

x + 64 = 289

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q1/5

1415

QUESTION ID

S46-B2-Q1

TYPE

Short answer

CORRECT ANSWER

225

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q1/5

Solve for x.

25 + x = 169

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q2/5

1416

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

144

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q2/5

Solve for x.

81 + x = 100

A. x = 19

B. x = 181

C. x = 9

D. x = 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q3/5

1417

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. x = 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q3/5

True or false?

x = 15 is the solution of x + 49 = 64.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q4/5

1418

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

x = 46 is the solution of 64 + x = 100.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q5/5

1419

QUESTION ID

S46-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The solution is x = 36.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 8.G.B.7

FORWARD 1

Solve right triangles with the Pythagorean theorem and trigonometric ratios • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A right triangle has a hypotenuse of 10 and a 30° angle. sin 30° = 0.5.

How long is the side opposite the 30° angle?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q1/5

1421

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q1**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q1/5

A ramp rises 3 feet over a horizontal distance of 4 feet.

How long is the ramp?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q2/5

1422

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q2/5

A kite string is 50 m long and makes a right triangle with the ground. The kite is 30 m above the ground.

How far is the kite horizontally from the person?

A. 58.3 m

B. 40 m

C. 20 m

D. 80 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q3/5

1423

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 40 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S46-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q3/5

True or false?

A right triangle with two 45° angles and legs of 7 has a hypotenuse of $7\sqrt{2}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q4/5

1424

QUESTION ID

S46-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q4/5

True or false?

tan 45° = 2

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q5/5

1425

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE tan 45° = 1.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 8.G.B.7

FORWARD 2

Apply geometric methods to solve design problems • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3

A rectangular field is 120 m by 50 m.

How much shorter is walking along the diagonal than walking along two sides?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q1/5

1427

QUESTION ID

S46-F2-Q1

TYPE

Short answer

CORRECT ANSWER

40 m (170 m compared with 130 m)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q1/5

A TV is sold as a 50-inch TV (its diagonal). It is 40 inches wide.

How tall is it?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q2/5

1428

QUESTION ID

S46-F2-Q2

TYPE

Short answer

CORRECT ANSWER

30 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q2/5

A cable runs from the top of a 12 m pole to a stake 5 m from the base of the pole.

How long is the cable?

A. 13 m

B. 17 m

C. √119 m

D. 7 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q3/5

1429

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 13 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q3/5

True or false?

A 10-foot ladder whose base is 6 feet from a wall reaches 8 feet up the wall.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q4/5

1430

QUESTION ID

S46-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q4/5

True or false?

The diagonal of a 9-by-12 rectangle is 21.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q5/5

1431

QUESTION ID

S46-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The diagonal is 15.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S46-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.3 • Set 46 • FORWARD 2 • Q5/5

SET 47 • 8.G.B.7 • GEOMETRY

Use the Pythagorean theorem to find unknown lengths in three dimensions

MAIN	Grade 8 main question	8.G.B.7
BACKWARD 1	Identify the faces, edges, and vertices of a prism	6.G.A.4
BACKWARD 2	Describe cross sections of right prisms	7.G.A.3
FORWARD 1	Use the Pythagorean theorem to solve problems in three dimensio	HSG-SRT.C.8
FORWARD 2	Identify solids made by rotating two-dimensional figures	HSG-GMD.B.4

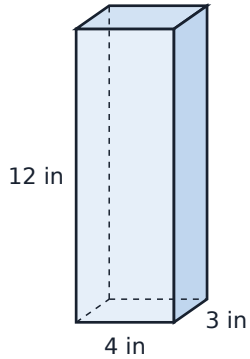
SET 47 • 8.G.B.7

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.7

A box is 4 inches long, 3 inches wide, and 12 inches tall.

How long is the diagonal from a bottom corner to the opposite top corner?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q1/5

1434

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

13 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q1/5

A cube has 6 cm edges.

Find the diagonal of one face, and then the diagonal from one corner of the cube to the opposite corner. Give exact values.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q2/5

1435

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

Face diagonal 6√2 cm; cube diagonal 6√3 cm

GRADING NOTE Grade both values.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q2/5

Will a 26-inch stick fit inside a box that is 24 by 6 by 8 inches, from one corner to the opposite corner?

A. No; the box is too short.

B. Yes; the face diagonal is 30 inches.

C. Yes; the corner-to-corner diagonal is exactly 26 inches.

D. No; the diagonal is only 24 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q3/5

1436

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Yes; the corner-to-corner diagonal is exactly 26 inches.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q3/5

A right square pyramid has its top directly above the center of its base. Its base edges are 10 m and its height is 12 m.

What is the slant height (from the top to the midpoint of a base edge)?

Slant height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q4/5

1437

QUESTION ID

S47-M-Q4

TYPE

Short answer

CORRECT ANSWER

13 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-M-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q4/5

True or false?

The corner-to-corner diagonal of a 2 by 3 by 6 box is 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q5/5

1438

QUESTION ID

S47-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 47 • MAIN • Q5/5

SET 47 • 8.G.B.7

BACKWARD 1

Identify the faces, edges, and vertices of a prism • Grade 6 • Geometry • 6.G.A.4

How many faces does a rectangular prism have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q1/5

1440

QUESTION ID

S47-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B1-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q1/5

How many edges does a rectangular prism have?

Edges: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q2/5

1441

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q2/5

How many vertices does a cube have?

A. 6

B. 4

C. 12

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q3/5

1442

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q3/5

True or false?

Each face of a rectangular prism is a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q4/5

1443

QUESTION ID

S47-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q4/5

True or false?

A triangular prism has 6 faces.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q5/5

1444

QUESTION ID

S47-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has 5 faces.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 8.G.B.7

BACKWARD 2

Describe cross sections of right prisms • Grade 7 • Geometry • 7.G.A.3

A right rectangular prism is sliced by a plane through two opposite vertical edges.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q1/5

1446

QUESTION ID

S47-B2-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q1/5

A cube is sliced by a plane parallel to one face, through its interior.

What is the shape of the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q2/5

1447

QUESTION ID

S47-B2-Q2

TYPE

Short answer

CORRECT ANSWER

A square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q2/5

A right rectangular prism is sliced parallel to its base, through its interior.

What is the cross section?

A. A circle

B. A triangle

C. A pentagon

D. A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q3/5

1448

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A rectangle the same size as the base

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-B2-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q3/5

True or false?

A diagonal slice through a box can make a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q4/5

1449

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q4/5

True or false?

A slice of a cube parallel to a face can be a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q5/5

1450

QUESTION ID

S47-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 8.G.B.7

FORWARD 1

Use the Pythagorean theorem to solve problems in three dimensions • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

A room is 12 ft long, 9 ft wide, and 8 ft high.

How long is the diagonal of the floor?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q1/5

1452

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

15 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q1**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q1/5

A room is 12 ft long, 9 ft wide, and 8 ft high.

How long is the diagonal from a floor corner to the opposite ceiling corner?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q2/5

1453

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

17 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q2/5

A box is 1 by 2 by 2.

How long is its corner-to-corner diagonal?

A. 3

B. 9

C. √5

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q3/5

1454

QUESTION ID

S47-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S47-F1-Q3**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q3/5

True or false?

The corner-to-corner diagonal of a cube with edge s is s√3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q4/5

1455

QUESTION ID

S47-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q4/5

True or false?

A box 3 by 4 by 5 has a corner-to-corner diagonal of 12.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q5/5

1456

QUESTION ID

S47-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is $\sqrt{50} = 5\sqrt{2}$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 8.G.B.7

FORWARD 2

Identify solids made by rotating two-dimensional figures • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4

A right triangle with legs 3 and 4 is rotated around the leg of length 4.

What solid does it make, and what is its slant height?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q1/5

1458

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

A cone with radius 3 and height 4; its slant height is 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q1/5

A 2-by-5 rectangle is rotated around one of its 5-unit sides.

What solid does it make?

Solid: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q2/5

1459

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

A cylinder with radius 2 and height 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q2/5

A sphere is sliced by a plane through its center.

What is the cross section?

A. A circle with the same radius as the sphere

B. An oval that is not a circle

C. A square

D. A smaller circle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q3/5

1460

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A circle with the same radius as the sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q3/5

True or false?

Rotating a semicircle about its diameter makes a sphere.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q4/5

1461

QUESTION ID

S47-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q4/5

True or false?

A cube can never have a triangular cross section.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q5/5

1462

QUESTION ID
S47-F2-Q5

TYPE
True or false

CORRECT ANSWER
B. False

GRADING NOTE A slice through three corners next to one vertex is a triangle.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S47-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.B.4 • Set 47 • FORWARD 2 • Q5/5

SET 48 • 8.G.B.8 • GEOMETRY

Find the distance between two points on the coordinate plane

MAIN	Grade 8 main question	**8.G.B.8**
BACKWARD 1	Find distances along grid lines	**6.NS.C.8**
BACKWARD 2	Find a distance as the absolute value of a difference	**7.NS.A.1.c**
FORWARD 1	Use the Pythagorean theorem with coordinates to solve problems	**HSG-SRT.C.8**
FORWARD 2	Use coordinates to find perimeters and areas of polygons	**HSG-GPE.B.7**

SET 48 • 8.G.B.8

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (7, 10).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q1/5

1465

QUESTION ID

S48-M-Q1

TYPE

Short answer

CORRECT ANSWER

10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q1/5

Find the distance between (−3, 4) and (2, −8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q2/5

1466

QUESTION ID

S48-M-Q2

TYPE

Short answer

CORRECT ANSWER

13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q2/5

What is the distance between (0, 0) and (4, 4), to the nearest tenth?

A. 16.0 units

B. 4.0 units

C. 5.7 units

D. 8.0 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q3/5

1467

QUESTION ID

S48-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5.7 units

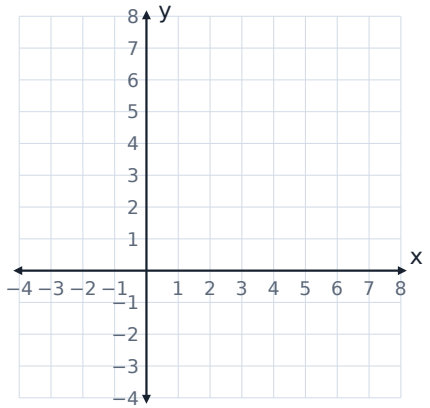
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q3/5

ANSWER KEY Reduced copy of the question page, with its answer

Plot A(−2, −1) and B(4, 7). Draw a right triangle with AB as its hypotenuse, and find AB.

AB = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q4/5

1468

QUESTION ID

S48-M-Q4

TYPE

Drawing + written answer

CORRECT ANSWER

Drawing: Points A and B, joined by the hypotenuse, with a horizontal leg of 6 units and a vertical leg of 8 units

Written: 10 units

GRADING NOTE Grade both: the right triangle with legs 6 and 8, and AB = 10 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-M-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q4/5

True or false?

The distance between (−1, −1) and (2, 3) is 5 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q5/5

1469

QUESTION ID

S48-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 48 • MAIN • Q5/5

SET 48 • 8.G.B.8

BACKWARD 1

Find distances along grid lines • Grade 6 • The Number System • 6.NS.C.8

Find the distance between (−4, 2) and (5, 2).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q1/5

1471

QUESTION ID

S48-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q1/5

Find the distance between (3, −6) and (3, 1).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q2/5

1472

QUESTION ID

S48-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q2/5

Which two points are 8 units apart?

A. (1, 1) and (1, 8)

B. (−3, 0) and (3, 0)

C. (−3, 0) and (5, 0)

D. (0, −4) and (0, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q3/5

1473

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. (−3, 0) and (5, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q3/5

True or false?

(2, −5) and (2, 5) are 10 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q4/5

1474

QUESTION ID

S48-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q4/5

True or false?

(−6, 3) and (−1, 3) are 7 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q5/5

1475

QUESTION ID

S48-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 8.G.B.8

BACKWARD 2

Find a distance as the absolute value of a difference • Grade 7 • The Number System • 7.NS.A.1.c

Find $|7 - (-2)|$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q1/5

1477

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q1/5

How far apart are −4 and 6 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q2/5

1478

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q2/5

Which expression gives the distance between −8 and −3?

A. |−8 − (−3)|

B. −8 − 3

C. |−8 + 3| − 1

D. −8 + (−3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q3/5

1479

QUESTION ID

S48-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. |−8 − (−3)|

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-B2-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q3/5

True or false?

The distance between −5 and 1 is 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q4/5

1480

QUESTION ID

S48-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q4/5

True or false?

The distance between −7 and −2 is 9.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q5/5

1481

QUESTION ID

S48-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 8.G.B.8

FORWARD 1

Use the Pythagorean theorem with coordinates to solve problems • High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8

On a map grid, a park is at (2, 3) and a school is at (10, 9). Each unit is 1 km.

How far apart are they in a straight line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q1/5

1483

QUESTION ID

S48-F1-Q1

TYPE

Short answer

CORRECT ANSWER

10 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q1

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q1/5

A drone flies from (0, 0) to (12, 5) on a grid in meters.

How far does it fly in a straight line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q2

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q2/5

1484

QUESTION ID

S48-F1-Q2

TYPE

Short answer

CORRECT ANSWER

13 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F1-Q2**

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q2/5

Which point is exactly 5 units from (1, 1)?

A. (6, 1)

B. (5, 5)

C. (1, 7)

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q3/5

1485

QUESTION ID

S48-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (4, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q3

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q3/5

True or false?

Every point that is 5 units from the origin lies on a circle with radius 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q4/5

1486

QUESTION ID

S48-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q4

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q4/5

True or false?

The points (0, 0) and (3, 3) are 6 units apart.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q5/5

1487

QUESTION ID

S48-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F1-Q5

High School (Grade 9 in this bank) • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.C.8 • Set 48 • FORWARD 1 • Q5/5

SET 48 • 8.G.B.8

FORWARD 2

Use coordinates to find perimeters and areas of polygons • High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7

Find the perimeter of the triangle with vertices (0, 0), (6, 0), and (6, 8).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q1

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q1/5

1489

QUESTION ID

S48-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q1/5

Find the area of the triangle with vertices (0, 0), (6, 0), and (6, 8).

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q2

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q2/5

1490

QUESTION ID

S48-F2-Q2

TYPE

Short answer

CORRECT ANSWER

24 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q2/5

A rhombus has vertices (0, 0), (3, 4), (8, 4), and (5, 0).

What is its perimeter?

A. 20 units

B. 18 units

C. 16 units

D. 25 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q3

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q3/5

1491

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q3**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q3/5

True or false?

A square with vertices (0, 0), (3, 1), (2, 4), and (−1, 3) has an area of 10 square units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q4/5

1492

QUESTION ID

S48-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q4

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q4/5

True or false?

The rectangle with vertices (0, 0), (4, 0), (4, 3), and (0, 3) has a perimeter of 12 units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S48-F2-Q5

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q5/5

1493

QUESTION ID

S48-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Its perimeter is 14 units.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S48-F2-Q5**

High School (Grade 10 in this bank) • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.7 • Set 48 • FORWARD 2 • Q5/5

SET 49 • 8.G.C.9 • GEOMETRY

Find the volume of a cylinder

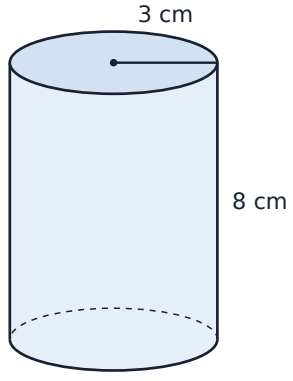
MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the area of a circle	**7.G.B.4**
BACKWARD 2	Find the volume of a right prism as base area times height	**7.G.B.6**
FORWARD 1	Give an informal argument for volume formulas	**HSG-GMD.A.1**
FORWARD 2	Use density with volume in a model	**HSG-MG.A.2**

SET 49 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the cylinder in terms of π.



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q1/5

1496

QUESTION ID

S49-M-Q1

TYPE

Short answer

CORRECT ANSWER

72π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q1/5

A cylinder has a diameter of 10 inches and a height of 6 inches.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q2/5

1497

QUESTION ID

S49-M-Q2

TYPE

Short answer

CORRECT ANSWER

471 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q2/5

A can holds 128π cubic centimeters and has a radius of 4 cm.

What is its height?

A. 8 cm

B. 2 cm

C. 16 cm

D. 32 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q3/5

1498

QUESTION ID

S49-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q3/5

A cylinder-shaped tank has a radius of 2 m and a height of 5 m.

How many liters does it hold? Use 3.14 for π. (1 cubic meter = 1,000 liters)

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q4/5

1499

QUESTION ID

S49-M-Q4

TYPE

Short answer

CORRECT ANSWER

62,800 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q4/5

True or false?

Doubling the radius of a cylinder doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q5/5

1500

QUESTION ID

S49-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It multiplies the volume by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 49 • MAIN • Q5/5

SET 49 • 8.G.C.9

BACKWARD 1

Find the area of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has a radius of 3.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q1/5

1502

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9π square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q1/5

A circle has a diameter of 10 cm.

What is its area? Use 3.14 for π.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q2/5

1503

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

78.5 square centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q2/5

A circle has a radius of 4 m.

What is its area in terms of π?

A. 64π square meters

B. 4π square meters

C. 16π square meters

D. 8π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q3/5

1504

QUESTION ID

S49-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q3/5

True or false?

A circle with a radius of 1 has an area of π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q4/5

1505

QUESTION ID

S49-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q4/5

True or false?

A circle with a diameter of 6 has an area of 36π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q5/5

1506

QUESTION ID

S49-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE The radius is 3, so the area is 9π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 8.G.C.9

BACKWARD 2

Find the volume of a right prism as base area times height • Grade 7 • Geometry • 7.G.B.6

A right prism has a base area of 20 square cm and a height of 7 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q1/5

1508

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

140 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-B2-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q1/5

A right triangular prism has a triangle base with area 6 square inches and a length of 10 inches.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q2/5

1509

QUESTION ID

S49-B2-Q2

TYPE

Short answer

CORRECT ANSWER

60 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q2/5

A right prism has a base area of 15 square m and a height of 4 m.

What is its volume?

A. 19 cubic meters

B. 30 cubic meters

C. 225 cubic meters

D. 60 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q3/5

1510

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q3/5

True or false?

The volume of any right prism is the area of its base times its height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q4/5

1511

QUESTION ID

S49-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q4/5

True or false?

A prism with a base area of 9 and a height of 2 has a volume of 11.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q5/5

1512

QUESTION ID

S49-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 49 • BACKWARD 2 • Q5/5

SET 49 • 8.G.C.9

FORWARD 1

Give an informal argument for volume formulas • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

Explain why the volume of a cylinder is the area of its base times its height.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q1/5

1514

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

The cylinder is a stack of equal circles (cross sections) with area πr^2; stacking them to height h gives $\pi r^2 \cdot h$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q1/5

A straight stack of coins and a slanted stack have the same coins and the same height.

Do they have the same volume? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q2/5

1515

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Yes. At every height the cross sections have the same area (Cavalieri's principle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q2/5

An oblique cylinder and a right cylinder have the same radius and height.

How do their volumes compare?

A. The oblique one is larger.

B. It cannot be known.

C. The right one is larger.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q3/5

1516

QUESTION ID

S49-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q3/5

True or false?

Cavalieri's principle compares the areas of cross sections at every height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q4/5

1517

QUESTION ID

S49-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q4/5

True or false?

Leaning a stack of paper to one side increases its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q5/5

1518

QUESTION ID

S49-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 49 • FORWARD 1 • Q5/5

SET 49 • 8.G.C.9

FORWARD 2

Use density with volume in a model • High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2

An aluminum cylinder has a radius of 2 cm and a height of 10 cm. Aluminum has a density of 2.7 grams per cubic centimeter.

What is its mass? Use 3.14 for π.

Mass: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q1/5

1520

QUESTION ID

S49-F2-Q1

TYPE

Short answer

CORRECT ANSWER

339.12 grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q1

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q1/5

A town has 12,000 people and covers 30 square miles.

What is its population density?

Density: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q2

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q2/5

1521

QUESTION ID

S49-F2-Q2

TYPE

Short answer

CORRECT ANSWER

400 people per square mile

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S49-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q2/5

A 500 cubic-centimeter block has a mass of 4,000 grams.

What is its density?

A. 8 grams per cubic centimeter

B. 2 grams per cubic centimeter

C. 0.125 gram per cubic centimeter

D. 4,500 grams

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q3/5

1522

QUESTION ID

S49-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 8 grams per cubic centimeter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q3

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q3/5

True or false?

If two objects have the same volume, the one with the greater density has the greater mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q4/5

1523

QUESTION ID

S49-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q4

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q4/5

True or false?

Density is volume divided by mass.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q5/5

1524

QUESTION ID

S49-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE Density is mass divided by volume.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S49-F2-Q5

High School (Grade 10 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.2 • Set 49 • FORWARD 2 • Q5/5

SET 50 • 8.G.C.9 • GEOMETRY

Find the volume of a cone

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the area of a circle from its radius or diameter	**7.G.B.4**
BACKWARD 2	Multiply a fraction by a whole number	**5.NF.B.4.a**
FORWARD 1	Explain why a cone's volume is one-third of a cylinder's	**HSG-GMD.A.1**
FORWARD 2	Use volume formulas to solve problems	**HSG-GMD.A.3**

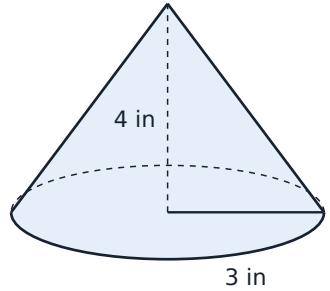
SET 50 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

Find the volume of the cone in terms of π.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q1/5

1527

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

12π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q1/5

A cone has a diameter of 12 feet and a height of 10 feet.

Find its volume. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q2/5

1528

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

376.8 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q2/5

A cone and a cylinder have the same base and the same height. The cylinder holds 90 cubic cm.

How much does the cone hold?

A. 30 cubic cm

B. 90 cubic cm

C. 270 cubic cm

D. 45 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q3/5

1529

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 30 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q3/5

A cone has a volume of 50π cubic units and a radius of 5 units.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q4/5

1530

QUESTION ID

S50-M-Q4

TYPE

Short answer

CORRECT ANSWER

6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q4/5

True or false?

A cone with a radius of 6 and a height of 2 has a volume of 24π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q5/5

1531

QUESTION ID

S50-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 50 • MAIN • Q5/5

SET 50 • 8.G.C.9

BACKWARD 1

Find the area of a circle from its radius or diameter • Grade 7 • Geometry • 7.G.B.4

A circle has a radius of 6.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q1/5

1533

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36π square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q1/5

A circle has a diameter of 4 m.

What is its area in terms of π?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q2/5

1534

QUESTION ID

S50-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4π square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q2/5

A circle has a radius of 10 cm.

What is its area? Use 3.14 for π.

A. 314 square cm

B. 31.4 square cm

C. 62.8 square cm

D. 628 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q3/5

1535

QUESTION ID

S50-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 314 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q3/5

True or false?

A circle with a radius of 2 has an area of 4π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q4/5

1536

QUESTION ID

S50-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q4/5

True or false?

A circle with a radius of 5 has an area of 10π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q5/5

1537

QUESTION ID

S50-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 8.G.C.9

BACKWARD 2

Multiply a fraction by a whole number • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{3} \times 36$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

1539

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q1

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q1/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{3} \times 90$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

1540

QUESTION ID

S50-B2-Q2

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q2

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Multiply.

$\frac{1}{3} \times 48$

A. 144

B. 12

C. 16

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

1541

QUESTION ID

S50-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q3

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q3/5

True or false?

$\frac{1}{3} \times 27 = 9$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

1542

QUESTION ID

S50-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-B2-Q4

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q4/5

True or false?

$\frac{1}{3} \times 15 = 45$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

1543

QUESTION ID

S50-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 8.G.C.9

FORWARD 1

Explain why a cone's volume is one-third of a cylinder's • High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1

A cone and a cylinder have the same base and height. Water from 3 full cones exactly fills the cylinder.

What formula does this suggest for the cone's volume?

Formula: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q1/5

1545

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$$V = \frac{1}{3}\pi r^2 h$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q1

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q1/5

Explain why an oblique cone and a right cone with the same base and height have the same volume.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q2/5

1546

QUESTION ID

S50-F1-Q2

TYPE

Short answer

CORRECT ANSWER

At every height their cross sections are circles with the same area (Cavalieri's principle).

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q2

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q2/5

A pyramid and a prism have the same base and height.

How do their volumes compare?

A. The pyramid's is $\frac{1}{3}$ of the prism's.

B. The prism's is $\frac{1}{3}$ of the pyramid's.

C. The pyramid's is $\frac{1}{2}$ of the prism's.

D. They are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q3/5

1547

QUESTION ID

S50-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. The pyramid's is 1/3 of the prism's.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q3

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q3/5

True or false?

A cone's volume formula has the same form as a pyramid's: $\frac{1}{3}$ × base area × height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q4/5

1548

QUESTION ID

S50-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q4

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q4/5

True or false?

A cone has half the volume of a cylinder with the same base and height.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q5/5

1549

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F1-Q5

High School (Grade 9 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.1 • Set 50 • FORWARD 1 • Q5/5

SET 50 • 8.G.C.9

FORWARD 2

Use volume formulas to solve problems • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A paper cone cup has a radius of 4 cm and a height of 9 cm.

How many full cups fill a cylinder-shaped jug with a radius of 8 cm and a height of 18 cm?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q1/5

1551

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q1/5

A cone-shaped pile of sand has a radius of 3 m and holds 30π cubic meters.

How tall is the pile?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q2/5

1552

QUESTION ID

S50-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S50-F2-Q2**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q2/5

A cone has a radius of 6 and a height of 8.

What is its slant height?

A. 14

B. 10

C. 48

D. √28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q3/5

1553

QUESTION ID

S50-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q3/5

True or false?

Doubling the height of a cone doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q4/5

1554

QUESTION ID

S50-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q4/5

True or false?

Doubling the radius of a cone doubles its volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q5/5

1555

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It multiplies the volume by 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S50-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 50 • FORWARD 2 • Q5/5

SET 51 • 8.G.C.9 • GEOMETRY

Find the volume of a sphere

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the volume of a cube	**6.G.A.2**
BACKWARD 2	Evaluate a formula with a fraction and an exponent	**6.EE.A.2.c**
FORWARD 1	Model objects with spheres, cylinders, and cones	**HSG-MG.A.1**
FORWARD 2	Use volume formulas to solve problems with spheres	**HSG-GMD.A.3**

SET 51 • 8.G.C.9

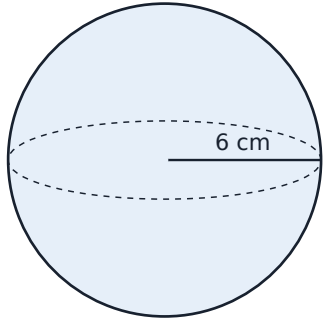
MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

ANSWER KEY Reduced copy of the question page, with its answer

Find the volume of the sphere in terms of π.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q1/5

1558

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

288π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q1/5

A ball has a diameter of 10 cm.

Find its volume to the nearest tenth. Use 3.14 for π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q2/5

1559

QUESTION ID

S51-M-Q2

TYPE

Short answer

CORRECT ANSWER

523.3 cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q2/5

A hemisphere (half of a sphere) has a radius of 3.

What is its volume in terms of π?

A. 36π

B. 18π

C. 9π

D. 27π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q3/5

1560

QUESTION ID

S51-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 18π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q3/5

A sphere has a volume of 36π cubic units.

What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q4/5

1561

QUESTION ID

S51-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q4/5

True or false?

A sphere with a radius of 1 has a volume of $\frac{4}{3}\pi$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q5/5

1562

QUESTION ID

S51-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 51 • MAIN • Q5/5

SET 51 • 8.G.C.9

BACKWARD 1

Find the volume of a cube • Grade 6 • Geometry • 6.G.A.2

A cube has edges of 3 units.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q1/5

1564

QUESTION ID

S51-B1-Q1

TYPE

Short answer

CORRECT ANSWER

27 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q1/5

A cube has edges of $\frac{1}{2}$ foot.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q2/5

1565

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/8 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q2/5

A cube has edges of 6 cm.

What is its volume?

A. 18 cubic cm

B. 216 cubic cm

C. 72 cubic cm

D. 36 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q3/5

1566

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 216 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B1-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q3/5

True or false?

A cube with 4-inch edges has a volume of 64 cubic inches.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q4/5

1567

QUESTION ID

S51-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q4/5

True or false?

A cube with 5-meter edges has a volume of 15 cubic meters.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q5/5

1568

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 8.G.C.9

BACKWARD 2

Evaluate a formula with a fraction and an exponent • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use $V = \frac{4}{3}r^3$. Find V when r = 3.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q1/5

1570

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q1/5

Use $V = \frac{4}{3}r^3$. Find V when r = 6.

V = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q2/5

1571

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

288

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q2/5

ANSWER KEY Reduced copy of the question page, with its answer

Use $V = \frac{1}{2}r^3$. Find V when r = 2.

A. 6

B. 4

C. 3

D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q3/5

1572

QUESTION ID

S51-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q3/5

True or false?

When r = 1, $\frac{4}{3}r^3 = \frac{4}{3}$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q4/5

1573

QUESTION ID

S51-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q4/5

ANSWER KEY Reduced copy of the question page, with its answer

True or false?

When r = 2, $\frac{4}{3}r^3 = 8$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q5/5

1574

QUESTION ID

S51-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It is 32/3.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 51 • BACKWARD 2 • Q5/5

SET 51 • 8.G.C.9

FORWARD 1

Model objects with spheres, cylinders, and cones • High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1

An orange is modeled as a sphere with a diameter of 8 cm.

Estimate its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q1/5

1576

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{256}{3}\pi$ cubic centimeters

GRADING NOTE About 268 cubic centimeters is also correct.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q1

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q1/5

A tennis ball can holds 3 balls in a cylinder whose radius equals the ball's radius r and whose height is 6r.

What fraction of the can is filled by the balls?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q2/5

1577

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q2

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q2/5

Which shape is the best model for a scoop of ice cream?

A. A cylinder

B. A sphere

C. A cone

D. A cube

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q3/5

1578

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. A sphere

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q3

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q3/5

True or false?

Modeling Earth as a sphere is a useful approximation even though Earth is slightly flattened.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q4/5

1579

QUESTION ID

S51-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q4

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q4/5

True or false?

A soccer ball is best modeled by a cylinder.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q5/5

1580

QUESTION ID

S51-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F1-Q5

High School (Grade 9 in this bank) • Geometry: Modeling with Geometry • HSG-MG.A.1 • Set 51 • FORWARD 1 • Q5/5

SET 51 • 8.G.C.9

FORWARD 2

Use volume formulas to solve problems with spheres • High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3

A spherical tank has a radius of 3 m.

How many liters does it hold? Use 3.14 for π. (1 cubic meter = 1,000 liters)

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q1

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q1/5

1582

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

113,040 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S51-F2-Q1**

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q1/5

A sphere and a cylinder have the same radius r, and the cylinder's height is 2r.

What is the ratio of the sphere's volume to the cylinder's volume?

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q2/5

1583

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2 : 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q2

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q2/5

Doubling a sphere's radius multiplies its volume by...

A. 6

B. 2

C. 8

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q3/5

1584

QUESTION ID

S51-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q3

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q3/5

True or false?

A sphere with volume 288π has a radius of 6.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q4/5

1585

QUESTION ID

S51-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q4

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q4/5

True or false?

A hemisphere with radius 6 has a volume of 288π.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q5/5

1586

QUESTION ID

S51-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE It has a volume of 144π.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S51-F2-Q5

High School (Grade 10 in this bank) • Geometry: Geometric Measurement & Dimension • HSG-GMD.A.3 • Set 51 • FORWARD 2 • Q5/5

SET 52 • 8.G.C.9 • GEOMETRY

Solve real-world problems with the volumes of cylinders, cones, and spheres

MAIN	Grade 8 main question	**8.G.C.9**
BACKWARD 1	Find the volume of a solid made of two prisms	**5.MD.C.5.c**
BACKWARD 2	Find the volume of a right prism	**7.G.B.6**
FORWARD 1	Model real objects with combinations of solids	**HSG-MG.A.1**
FORWARD 2	Apply volume to design and capacity problems	**HSG-MG.A.3**

SET 52 • 8.G.C.9

MAIN

Grade 8 main question • Grade 8 • Geometry • 8.G.C.9

A grain silo is a cylinder with a radius of 6 m and a height of 10 m, topped with half of a sphere with a radius of 6 m.

Find its volume in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q1/5

1589

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

504π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q1/5

An ice cream cone (radius 3 cm, height 10 cm) is filled and topped with half of a sphere of ice cream with a radius of 3 cm.

Find the total volume of ice cream in terms of π.

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q2/5

1590

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

48π cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q2/5

A cone and a cylinder each have a radius of 2 and a height of 6.

How many full cones of water fill the cylinder?

A. 6

B. 3

C. 1

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q3/5

1591

QUESTION ID

S52-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q3/5

A cylinder with a radius of 5 cm holds water. A ball with a radius of 3 cm is dropped in and sinks completely.

How many centimeters does the water level rise?

Rise: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q4/5

1592

QUESTION ID

S52-M-Q4

TYPE

Short answer

CORRECT ANSWER

1.44 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q4/5

True or false?

A sphere with radius r fits exactly in a cylinder with radius r and height 2r. The sphere's volume is $\frac{2}{3}$ of the cylinder's.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q5/5

1593

QUESTION ID

S52-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-M-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 52 • MAIN • Q5/5

SET 52 • 8.G.C.9

BACKWARD 1

Find the volume of a solid made of two prisms • Grade 5 • Measurement & Data • 5.MD.C.5.c

A 4 by 3 by 2 prism and a 2 by 3 by 2 prism are joined with no overlap.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q1/5

1595

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q1/5

Two cubes with 2-unit edges are joined face to face.

What is the total volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q2/5

1596

QUESTION ID

S52-B1-Q2

TYPE

Short answer

CORRECT ANSWER

16 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q2/5

A 5 by 2 by 1 prism and a 1 by 2 by 1 prism are joined with no overlap.

What is the total volume?

A. 12 cubic units

B. 7 cubic units

C. 20 cubic units

D. 10 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q3/5

1597

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 12 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q3/5

True or false?

Volumes of parts that do not overlap can be added to find a total volume.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q4/5

1598

QUESTION ID

S52-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q4/5

True or false?

A 3 by 3 by 3 cube and a 1 by 1 by 1 cube joined with no overlap have a volume of 30 cubic units.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q5/5

1599

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

GRADING NOTE 27 + 1 = 28.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S52-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.c • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 8.G.C.9

BACKWARD 2

Find the volume of a right prism • Grade 7 • Geometry • 7.G.B.6